University of Amsterdam

Master's Thesis

Asymmetric Belief Networks
Inference and Intervention

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in the

Computational Science Lab
Informatics Institute

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Declaration of Authorship

I, Henry ZWART, declare that this thesis, entitled ‘Asymmetric Belief Networks Inference and Intervention’ and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at the University of Amsterdam.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed:

Date: 22 June 2026

"*What I cannot create, I do not understand.*"

— Richard P. Feynman

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Abstract

Faculty of Science
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Master of Science in Computational Science

Asymmetric Belief Networks
Inference and Intervention

by Henry ZWART

The relationship between individual preferences and behaviour is well-entrenched in common knowledge, and is core to most theories of behaviour. It follows that interventions at the level of individuals' beliefs, preferences, and attitudes (henceforth 'belief') may be an effective vector for broader behavioural change. However, interventions on internal states such as these are necessarily indirect. Moreover, these beliefs do not exist in isolation, so interventions may have intended consequences for tangential beliefs. While recent studies (citation needed) have investigated correlational relations between beliefs, relatively little consideration

Acknowledgements

Thank the people that have helped: supervisors, family, etc.

Use of AI

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Contents

Bibliography **65**

List of Figures

List of Tables

List of Algorithms

Abbreviations

CSL **C**omputational **S**cience **L**ab

UvA **U**niversiteit **v**an **A**msterdam

Chapter 1

Notes (things to remember, to-dos)

See (Epskamp et al., 2018) for discussion on the use of bootstrapping in assessing edge weight variability. Also argument against the use of bootstrap CIs to test for edges being non-zero; particularly in LASSO-regularised models.

The general Markov description of belief system dynamics in methods doesn't assume that belief/attitude states are numerical or even 1-dimensional. But we do assume this in our model.

Chapter 2

Introduction

Old RQs:

- How do beliefs about climate change, and the relations between beliefs, vary across population groups and under different structural and causal assumptions regarding belief dynamics?

- To what degree are the implications (outcome, effectiveness) of belief-level interventions influenced by structural and causal assumptions regarding belief dynamics?

Revised (potential) research foci:

Theoretical contributions:

- **RF1.1:** Extending the causal attitude network model of belief systems to support asymmetric causal effects between beliefs and attitudes.

- **RQ1.1:** What is the representational capacity of a pairwise belief system model?

- **RQ1.2:** How do belief system dynamics differ between models assuming symmetric relations, and those assuming asymmetric relations? If the asymmetric model reaches a steady state, is this an equilibrium state? Should we expect the steady state to be similar to the ESS of the symmetric model? Under what conditions do we expect the asymmetric model to converge to a steady state monotonically? How does the steady state depend on the initial conditions of the belief system?

- **RQ1.3:** How is the process of inferring belief systems from observational data sensitive to unmeasured factors, or incorrect structural assumptions? What is the impact of excluding true relations — or including false ones — on an inferred model?

Inferring belief systems:

- **RQ2.1:** How do (symmetric or asymmetric) belief systems relating to climate change vary between individuals with different background contexts, as inferred from observational data?

- **RQ2.2:** To what extent are causal relations *asymmetric* in models of climate change belief systems inferred from observational data?

Intervention dynamics:

- **RQ3.1:** How do intervention strategy and effectiveness differ between symmetric and asymmetric belief system models for a given observational dataset?

- **RQ3.2:** How do intervention outcome and effectiveness vary between individuals with different initial conditions, or with different backgrounds?

Chapter 3

Literature review

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Chapter 4

Climate beliefs dataset

- Longitudinal survey overview
- Question selection
- Cleaning and transformation
- Validation
- Imputation

4.1 Longitudinal climate attitudes survey

Introduce the survey:

- Motivate it: This is the main resource used to construct our dataset.
- Description:
  - Content: What sorts of questions? (Examples)
  - Waves: how many, when were they run? (Participation event plot)
  - Participation: counts, variability. (Statistics: how many people answered all waves? How many singletons?)
  - Question variability. (…?)
- Overlap with notable events:
  - COVID-19 outbreaks
  - US electoral cycle
  - Major weather events

The longitudinal study (**CITE**) comprises six waves of responses from individuals residing in the United States, collected between April 2020 and March 2023. The assessed dimensions include general demographic information (e.g., age, gender, education, and financial status), beliefs, attitudes, and experiences relating to concurrently-salient

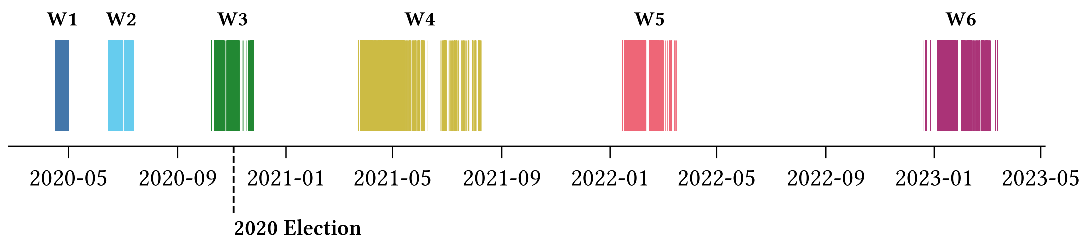

Figure 4.1: Longitudinal survey response dates per-wave.

topics such as COVID-19, climate change, or the 2020 US presidential election, and support for hypothetical policies.

- Significance of longitudinal data — most studies use cross-sectional data which has limited interpretability at individual level.

Survey content; nature of the questions asked

- Context
- Category (belief, attitude, demographic, etc.)
- Response formats

The survey content varies between waves, and between participants.

Participation

Waves The survey waves occur with irregular spacing and duration, as illustrated in Figure 4.1. Some consecutive pairs of waves are much closer than others. For instance, the joined interval spanned by the first three waves has a longer duration than the interval between Waves 5 and 6, and the duration spanned by Wave 4 exceeds that of the joined interval comprising Waves 1 and 2.

The 2020 US Presidential Election date occurred during Wave 3. This wave was modified for responses started after the election date; future-focused survey questions regarding individuals' voting intentions, beliefs, and attitudes concerning the election were removed.

- Context of the survey:
  - Time-span
  - Wave timing/duration
  - Overlap with COVID-19 outbreaks and US electoral cycle, major weather events
- Codebook covers first five waves
- Nature of the questions asked:
- Variability in questions (between waves, participants)

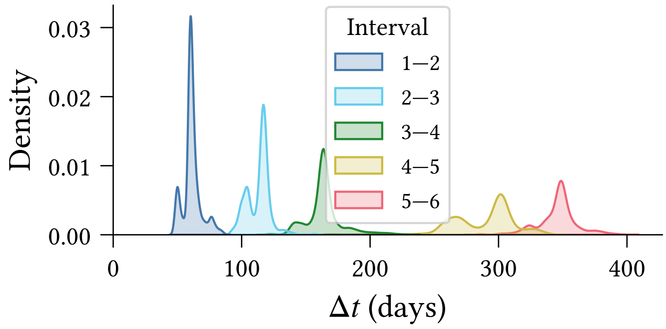

Figure 4.2: Distribution of inter-response times across individuals.

4.2 Question selection

- Motivation (identify measures of distinct beliefs and attitudes regarding climate change – impacts, risks, policies, etc.)
- Criteria for selecting questions
- Exclude sixth wave due to no codebook

4.3 Validation

- Only validate selected questions
- Validation components:
  - Schema (Pandera)
  - Null responses iff null expected
  - Valid survey responses
  - Miscellaneous inconsistencies
    - Born in US in some waves, not in others
    - Changing responses to extreme weather experience in last 10 years
- Results:
  - Schema good
  - Various issues with Null responses:
    - Some due to survey logic errors
    - Some not explained

4.4 Normalisation, cleaning, and transformations

- Variable names (replace '.' with double underscore)
- Filter out participants with:
  - Null `participant_id`

  - Survey errors (from validation above)
  - Failed survey validation checks (if any check is failed, participant is removed entirely)
- Response type/value conversions
  - Non-float numeric variables to Int64
  - Re-map $1.. = k$ variables to $(0.. = k - 1)$
  - Replace empty string entries with `Null`
  - Multiple choice questions: comma-separated string of integers to list of integers/enums
  - Replace categorical variable integer responses with enums (for human-readability and to ensure value consistency across waves)
  - `cc2`: Re-order such that human-causes are 'high values' and natural-causes/no climate change are 'low values'
- Coalescing treatment columns:
  - Responses stored in one column
  - Separate index column specifies treatment group membership
  - Optional third column for treatment-specific numeric parameters (e.g., policy cost)
- Constructed columns:
  - Merge `pol_party` and `pol_lean` to create five-point `pol_affiliation` scale. Note that this does not capture individuals who 'lean *right* toward Democrat' for example.

4.5 Missing-wave imputation

- Survey responses as Markov processes
- Viterbi imputation:
  - Maximum likelihood estimation based on transition matrix
  - Independence assumption (assumes variables are independent in missing waves), can only reduce measured dependence between variables.
    - Maybe I can give an info theory proof of this?
  - Allows us to consider questions which are not always asked in same waves.
- Evaluation:
  - Look at imputation impacts on manually degraded dataset, perhaps from COVID questions (which I'm not using, but which are analogous to some of our key questions).
  - How is the direct performance on individual variables? How does it degrade with the number of imputed waves, or with the locations of the non-null waves?
  - How does dependence/correlation between related variables degrade as we increase the number of imputation waves?

4.6 Data Validation

The complexity of the climate attitudes survey (**reference section discussing this**) necessitates a rigorous approach to data validation, both to identify errors in the expected schema and to ensure that the data matches our expectations.

We have implemented a general validation pipeline comprising three stages:

1. **Type-level validation:** The set of columns is exactly as expected, and all columns have the correct data type.

2. **Response-value validation:** All *non-null* responses are valid according to the schema specified in the codebook.

3. **Null-value validation:** Responses are null if, and only if, we expect them to be null.

In the first instance, type-level validation ensures that the *observed* data schema matches the *prescribed* schema. Since response data types vary between questions according to the question type (see Table 4.1), the type-checking must be flexible and capable of handling complex data types.

For instance, categorical multiple-choice responses are represented using a `list[Enum]`, where `Enum` is a question-specific enum-type[1]. Type validation for a multiple-choice question thus requires checking that the response column comprises `list`'s whose elements are all from the set of values defined by the `Enum`.

Response-value validation then ensures that all non-null responses are valid according to the set of allowable values described in the survey codebook. For most variables this is straightforward. Numeric and single-response ordinal variables typically have

Response schema	Raw type	Coerced type
Text	`str`	`str`
Single response (ordinal)	`int`	`int`
Single response (categorical)	`int`	`Enum`
Multiple response	`str`	`list[Enum]`
Numeric	`float`	`float | int`

Table 4.1: **TODO**

[1]An enum is a type defined by a finite set of allowable values. In our case the values are human-readable strings. For instance, the `dem_urban` survey question, which asks 'What kind of area do you live in?' has responses with data type described by the enum $\{\text{Urban}, \text{Suburban}, \text{Rural}\}$.

a defined range (e.g., $18 \leq \text{age} \leq 99$, or $1 \leq x \leq 5$ for a 5-point Likert scale variable). Text-entry responses are always considered valid.

Single-response categorical questions have `Enum` type, so type-level validation is sufficient to ensure that the responses do not contain any values outside the set defined by the `Enum`. However, the allowable values occasionally change between survey waves. Hence response-value validation is also required for categorical single-response and multiple-response variables, and in general must handle with schema variation between waves.

We implement both type-level and response-value validation using the Pandera Python library for DataFrame validation (Bantilan et al., 2022). The Pandera API allows for straightforward schema specification using Python type annotations (for type-level validation) in combination with value constraints (for response-level validation). We implement a separate validation check for questions whose response schema varies between survey waves, using a Pandera 'dataframe check' to stratify validation by wave.

Null-value validation ensures that responses are null if, and only if, they are expected to be null. For a question Q, the response of a participant P in wave W can be null for one of four reasons:

N1. Q is not asked in wave W,

N2. In wave W, Q is only shown to *new* participants, but P is a *repeating* participant (or vice versa),

N3. In wave W, Q is only shown to a treatment group of which P is not a member, or

N4. In wave W, Q is conditionally shown to participants based on their response to a distinct question Q', and P's response to Q' does not satisfy the condition.

Formally we can write the condition for P's response R being null as:

$$\text{null}(R) \iff (\text{N1} \lor \text{N2} \lor \text{N3} \lor \text{N4}) \tag{4.1}$$

We split this up into: If not in wave, then R is null.

$$(\text{N1} \lor \text{N2}) \implies \text{null}(R) \tag{4.2}$$

If conditions not met, then R is null.

$$(\text{N3} \lor \text{N4}) \implies \text{null}(R) \tag{4.3}$$

For unconditional columns, if shown, then not null:

$$\neg(\mathrm{N1} \lor \mathrm{N2}) \implies \neg\,\mathrm{null}(R) \tag{4.4}$$

For conditional columns, if shown, then not null:

$$\neg(\mathrm{N1} \lor \mathrm{N2} \lor \mathrm{N3} \lor \mathrm{N4}) \implies \neg\,\mathrm{null}(R) \tag{4.5}$$

For a conditional $A \implies B$, let $\boldsymbol{R}$ be the subset of responses for which the contradiction $\neg(A \implies B) \equiv (A \land \neg B)$ is true. Then $A \implies B$ is satisfied if, and only if, $\boldsymbol{R}$ is empty **check this — is it iff, if, or only if?**. If $\boldsymbol{R}$ is non-empty, then the contained responses are counter-examples to the conditional.

- Single-response ordinal:

In the first case, type-level validation ensures that the data schema matches expectations, independently of the response values. In the first case, type-level validation ensures that the schema observed in the dataset matches the schema Type-level validation Our validation pipeline comprises three stages:

The first two stages are handled by Pandera, and the third is manually implemented.

- What does type-level validation consist in?
  - Simple types
  - Enums
  - Lists of enums
- What does response value validation consist in?
  - Checking that observed (non-null) values are within the set of allowable values
- What does null-value validation consist in?
  - ...

The data schema defines everything that is necessary. Maybe we can discuss this separately as an implementation step.

Chapter 5

Terminology and notation

- $\langle ij \rangle$: The set of interacting pairs of spins.
- $[N]$: The index set $1.. = N$.
- log: Base 2 unless otherwise stated.

Chapter 6

Methods

6.1 Belief system dynamics

- Interdependent beliefs and attitudes; state of one affects the state of another:
  - Cognitive dissonance
  - Causal effects(?)
- Individual belief systems:
  - Degree of influence depends on individual experience, perception, meta-level beliefs.
- Formalising dynamics:
  - Discrete beliefs, attitudes which take on values in some domain
  - State of each one depends on previous in a Markov process
  - Broader Markov process describes the state of the belief system as a whole

We will now formalise the dynamics of an individual belief system, under the following assumptions:

1. Beliefs and attitudes can be treated as discrete entities.

2. The state of a belief or attitude is Markovian with respect to the previous system configurations.

3. The probability distribution relating current and future belief states does not change over time.

The first assumption includes two key points. Firstly, that it makes sense to consider beliefs and attitudes as *entities*, or *objects*. This contrasts, for instance, alternative interpretations of attitudes as emissions from an underlying structure. (**TODO:** review CAN paper for further discussion on this matter). Secondly, that beliefs and attitudes can be considered *discrete* or *separable*. This is to say that there are no absolute physical constraints on the combinations of states which may be observed (although certain combinations may be highly unlikely). In particular, this captures the requirement that each belief or attitude can be represented using a single state variable.

Our second assumption concerns the treatment of beliefs and attitudes as Markovian.

- Harder to justify conceptually, since individuals have memory and can recall prior states beyond the previous second.

Consider a belief or attitude S_i with domain Ω_{S_i}. The state of S_i at time t depends on the previous states of other beliefs and attitudes, and we can therefore describe the trajectory of S_i as a Markov process:

$$\{s_i^t\}_{t=0}^{\infty} \quad \text{where} \quad s_i^t \in \Omega_{S_i} \tag{6.6}$$

with conditional state probability $P\left(S_i^{\tau+1} \mid \boldsymbol{S}^1, ..., \boldsymbol{S}^\tau\right) = P\left(S_i^{\tau+1} \mid \boldsymbol{S}^\tau\right)$ for $\tau \in \mathbb{N}$.

Since the state of S_i^{t+1} depends only on the previous belief system state, it follows that for any other belief or attitude S_j, the states S_i^{t+1} and S_j^{t+1} are conditionally independent given $\boldsymbol{S}^t$. We can therefore straightforwardly extend Equation 6.6 to describe the evolution of the entire belief system as a Markov process,

$$\{\boldsymbol{s}^t\}_{t=0}^{\infty} \quad \text{where } \boldsymbol{s}^t \in \Omega_{S_1} \times \cdots \times \Omega_{S_n} \tag{6.7}$$

with the probability of each belief system configuration given by the conditional distribution $P(\boldsymbol{S}^{t+1} \mid \boldsymbol{S}^t)$.

Under this conceptualisation, the task of modelling a belief system thus reduces to describing how the instantaneous configuration of belief states affects the probability distribution over possible future states.

6.2 Asymmetric belief system model

6.3 Experimental plan

6.4 Dataset

6.4.1 Variable selection

6.4.2 Binarisation

Prior to model fitting, we binarise all variables, mapping data values to the domain $\{-1, +1\}$. Before binarising, we shift each variable such that the *survey midpoint* (e.g., '3' on a 5-point Likert scale) aligns with 0, and rescale each variable such that the minimum and maximum possible values map to -1 and $+1$ respectively. For variables with no well-defined midpoint, such as those with an even number of possible responses, we shift such that the minimal and maximal values are at equal distance from 0.

We use a smooth binarisation process to robustly handle data points which are zero, or close to zero. This involves perturbing each data value $x \in \mathbb{R}$ by an independent noise term $\varepsilon \sim \mathcal{N}(0, \sigma)$ before thresholding. Figure 6.3 illustrates this process for a negative value of x and given choice of σ.

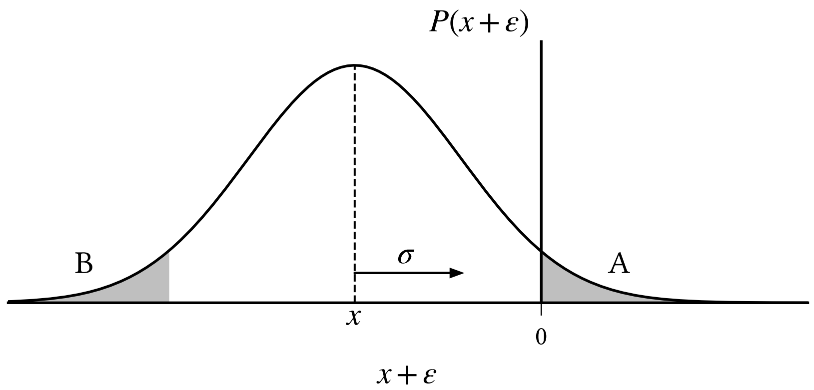

Figure 6.3: $x \in \mathbb{R}$ is smoothly binarised to $\{-1, +1\}$ by thresholding $x' = (x + \varepsilon)$ where $(x + \varepsilon) \sim \mathcal{N}(x, \sigma)$. Negative values are mapped to $+1$ with probability $P(x' > 0) = \mathrm{A} = \mathrm{B} = P(\varepsilon < x)$, which increases with σ and $|x|^{-1}$.

Observe that x is mapped to $+1$ if, and only if, ε is sufficiently large, such that $x + \varepsilon > 0$ (region A), or equivalently when $\varepsilon < x$ (region B). The probability that x is mapped to $+1$ is then

$$P(x \mapsto +1) = \Phi\left(\frac{x}{\sigma}\right) \tag{6.8}$$

where Φ is the standardised normal cumulative distribution function. This probability is small when x has large magnitude, or when σ is small. For the purposes of our experiments, we choose $\sigma = 0.2$ such that a ‘weakly oppose’ response to a 7-point Likert scale[2] (value 1/3) is mapped to +1 with probability 0.05.

6.5 Parameter recovery

6.5.1 Regularisation

Both the symmetric and asymmetric belief system models contain a relatively large number of parameters to be estimated compared to the number of observations in the dataset, which poses an issue for obtaining reliable parameter estimates (Epskamp et al., 2018). In line with standard practice, we apply regularisation during estimation to encourage model sparsity, using a variation of LASSO regularisation (Tibshirani, 1996) for which the first derivative is smooth.

LASSO regularisation (**l**east **a**bsolute **s**hrinkage and **s**election **o**perator) is implemented as an additional summand in the objective function. Rather than maximising the log-likelihood directly, we choose parameters $\hat{\boldsymbol{\theta}} \in \mathbb{R}^p$ such that

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}' \in \mathbb{R}^p}{\operatorname{argmax}} \left\{ \mathcal{L}_D(\boldsymbol{\theta}') - \lambda \sum_{i=1}^{p} |\theta'_i| \right\} \tag{6.9}$$

This has the effect of penalising non-zero parameters with negligible contribution to the log-likelihood. The parameter $\lambda \in \mathbb{R}^+$ controls regularisation strength, with larger values resulting in sparser models.

This formulation of LASSO regularisation has a discontinuous first-derivative at $\theta'_i = 0$ due to the absolute value function. Instead, we use the following smooth variation, where $\varepsilon \in \mathbb{R}^+$:

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}' \in \mathbb{R}^p}{\operatorname{argmax}} \left\{ \mathcal{L}_D(\boldsymbol{\theta}') - \lambda \sum_{i=1}^{p} \sqrt{\theta_i'^2 + \varepsilon} \right\} \tag{6.10}$$

[2]The oppose/support 7-point Likert scale has possible responses: strongly oppose, oppose, weakly oppose, neutral, weakly support, support, strongly support.

Equation 6.10 converges to Equation 6.9 as $\varepsilon \to 0^+$, and has a well-defined first-derivative for $\varepsilon > 0$, namely:

$$\frac{\mathrm{d}}{\mathrm{d}\theta'_i}\left(-\lambda \sum_{j=1}^{p} \sqrt{\theta'^2_j + \varepsilon}\right) = \frac{\lambda \theta'_i}{\sqrt{\theta'^2_i + \varepsilon}} \tag{6.11}$$

For the purposes of this study, we will consider parameters as 'effectively non-zero' if their magnitude exceeds 10^{-2}. We thus choose $\varepsilon = 10^{-8}$, such that $\sqrt{\varepsilon}$ is two orders of magnitude smaller than the minimal parameter size of interest, ensuring that Equation 6.10 is a reasonable approximation to Equation 6.9 for parameter values smaller than 10^{-2}.

- Effects: How does regularisation affect sparsity?

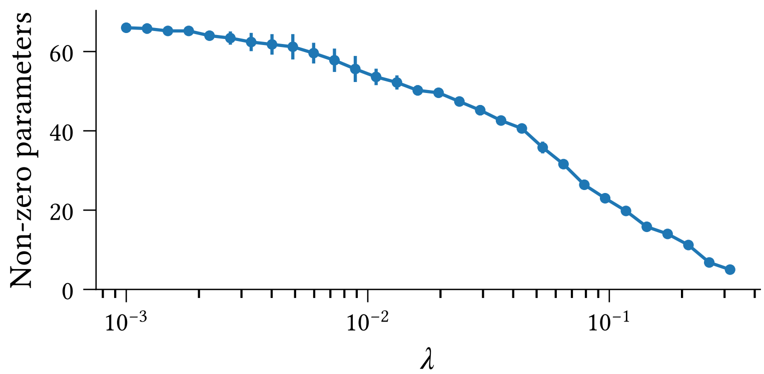

Figure 6.4: Model sparsity increases with regularisation strength (λ). Error bars display 95% confidence intervals around expected number of non-zero parameters, measured across different stochastic binarisations of observed data.

We use the Extended Bayesian Information Criterion (EBIC, Equation 6.12) (Chen & Chen, 2008) to select an appropriate regularisation strength, as recommended in Epskamp et al. (2018). The EBIC is an evaluative criterion for model selection, which balances maximisation of the log-likelihood with model complexity, as measured by the number of parameters. Compared to the original Bayesian Information Criterion, EBIC tends to be more conservative when the number of parameters and observatons are comparable (Foygel & Drton, 2010).

For a model with p parameters, k of which are non-zero, fit using a dataset with n observations, the EBIC is defined as:

$$\mathrm{EBIC}(\boldsymbol{\theta}) = k \cdot [\ln(n) + 2\gamma \ln(p)] - 2\mathcal{L}_D(\boldsymbol{\theta}) \tag{6.12}$$

We take $\gamma = 0.25$, which has been shown to work well for network inference in the inverse Ising problem (Barber & Drton, 2015).

In general, models with a smaller EBIC are preferred, as this indicates a reasonable trade-off between model fit and complexity. We fit each of the symmetric and asymmetric models for candidate regularisation strengths $\lambda \in \mathbb{R} \subset [10^{-4}, 10^{-1}]$, and select the values for which the resulting EBIC is minimal (Figure 6.5). This yields symmetric and asymmetric regularisation strengths of 0.006 and 0.009 respectively.

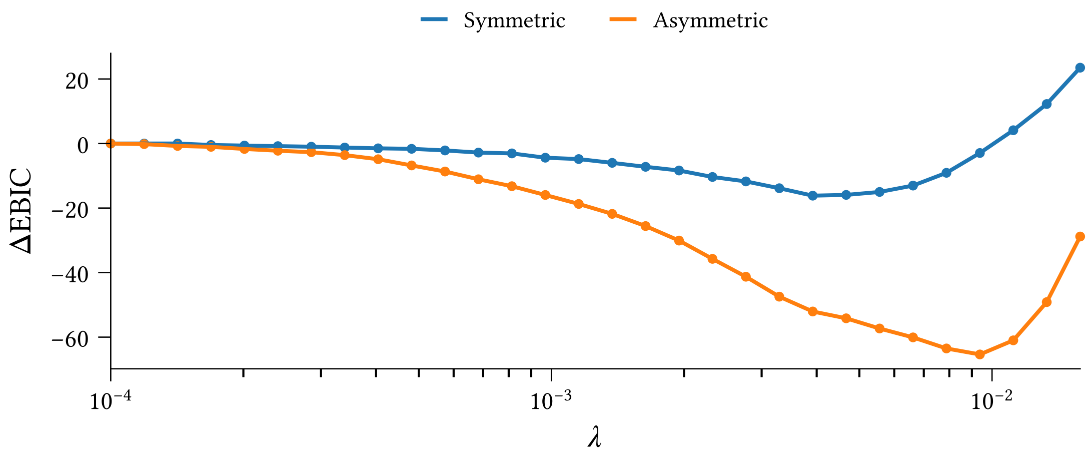

Figure 6.5: **TODO**

6.6 Toy models (?)

Chapter 7

Asymmetric belief system model

$$s^{t+1} \sim P(\boldsymbol{S}^{t+1}|\boldsymbol{S}^{t}) \tag{7.13}$$

7.1 Summary

In the following pages I summarise the current state of the project and our intentions going forward. I begin by outlining the (revised) research questions, followed by our tentative experimental plan for answering these questions. Finally I provide theoretical details regarding the asymmetric belief system model, including simulation and intervention dynamics, and methods for inferring belief systems from observational data.

7.1.1 Research questions

Theoretical contributions:

- **RF1.1:** Extending the causal attitude network model of belief systems to support asymmetric causal effects between beliefs and attitudes.
- **RQ1.1:** What is the representational capacity of a pairwise belief system model?
- **RQ1.2:** How do belief system dynamics differ between models assuming symmetric relations, and those assuming asymmetric relations? If the asymmetric model reaches a steady state, is this an equilibrium state? Should we expect the steady state to be similar to the ESS of the symmetric model? Under what conditions do we expect the asymmetric model to converge to a steady state monotonically? How does the steady state depend on the initial conditions of the belief system?

- **RQ1.3:** How is the process of inferring belief systems from observational data sensitive to unmeasured factors, or incorrect structural assumptions? What is the impact of excluding true relations — or including false ones — on an inferred model?

Inferring belief systems:

- **RQ2.1:** How do (symmetric or asymmetric) belief systems relating to climate change vary between individuals with different background contexts, as inferred from observational data?

- **RQ2.2:** To what extent are causal relations *symmetric* or *asymmetric* in models of climate change belief systems inferred from observational data?

Intervention dynamics:

- **RQ3.1:** How do intervention strategy and effectiveness differ between symmetric and asymmetric belief system models for a given observational dataset?

- **RQ3.2:** How do intervention outcome and effectiveness vary between individuals with different initial conditions, or with different backgrounds?

7.1.2 Experiment plan

IN PROGRESS

7.1.2.1 Binarisation of survey data

We have constructed a reduced version of the climate attitudes dataset, comprising a restricted set of columns relating to climate change beliefs and attitudes, as well as a small number of index variables. Each column in the dataset is centered such that the survey-midpoint for the associated question (e.g., "maybe", "don't know", or "neither support nor oppose") is mapped to zero. Index columns are centered prior to index calculation.

As the belief system model described in Chapter 7.1.3 assumes belief configurations in $\{0,1\}^N$, binarisation is required before these models can be fit to the survey data. We binarise the data with noise to smooth the transition around the ambiguous zero-state.

Let $D = \mathbb{R}^{M\times t\times N}$ be a dataset comprising observations from $M \in \mathbb{N}$ individuals at $t \in \mathbb{N}$ timepoints, for a belief system with $N \in \mathbb{N}$ items. The binarisation process can be viewed as sampling a binary dataset D_b with:

$$\left(D_b\right)_{i,j} = \begin{cases} +1 & \tilde{D}_{i,j} \geq 0 \\ -1 & \tilde{D}_{i,j} < 0 \end{cases} \tag{7.14.1}$$

$$\tilde{D}_{i,j} \sim \mathrm{Normal}\big(D_{i,j}, \sigma\big) \tag{7.14.2}$$

where $\sigma \in \mathbb{R}_{\geq 0}$ controls the level of stochasticity, and the resulting sharpness in the step from -1 to $+1$. Whenever multiple repeats are performed for a given experiment, we re-binarise the dataset for each repeat.

7.1.3 Asymmetric belief systems

Let S be a set containing $N \in \mathbb{N}$ cognitive axes (beliefs or attitudes). Adopting Ising-model terminology, we refer to the elements of S as **spins**. We define the **observed states** of a spin $S_i \in S$ as a Markov process

$$\left\{s_i^t\right\}_{t=0}^{\infty} \quad \text{where} \quad s_i^t \in \{-1, +1\} \tag{7.15}$$

and the **observed configurations** of S as

$$\left\{\boldsymbol{s}^t\right\}_{t=0}^{\infty} \quad \text{where} \quad \boldsymbol{s}^t = \left[s_1^t, ..., s_N^t\right]^T \in \{-1, +1\}^n \tag{7.16}$$

For $n \in \mathbb{N}_{>0}$ we denote the distribution over observed states for a spin $S_i \in S$ by $P_{S_i^n|S_i^{n-1}}$ and the distribution over observed configurations as $P_{\boldsymbol{S}^n|\boldsymbol{S}^{n-1}}$.

We define a **pairwise-interaction belief network** over the elements in S as a directed network $G = \langle S, \boldsymbol{A} \rangle$, such that S forms to set of nodes in G, and $\boldsymbol{A} \in \{0, 1\}^{N \times N}$ is a directed adjacency matrix. For $s_i, s_j \in S$, we have $A_{ij} = 1$ if, and only if, the distribution of observed states for S_j^n depends on the previous observed state of S_i^{n-1}:

$$P_{S_j^n|\boldsymbol{S}^{n-1}} \neq P_{S_j^n|\boldsymbol{S}_{-i}^{n-1}} \text{ for some } n \in \mathbb{N}_{>0} \tag{7.17}$$

In such cases we say that S_i *directly influences* S_j, and write $S_i \to S_j$.

Before proceeding, it is important to clarify the use of the term ‘belief system’ in this thesis, particularly in the context of populations of individuals. The notion of a belief system is an inherently individual one. Indeed, the behaviour of, and interactions between, any individual’s beliefs and attitudes is necessarily shaped by their distinct context in the world. In spite of this, we will often speak of belief systems as if belonging to a population. It is important to stress that in this context, the term ‘belief system’ rather refers to the family of individual belief systems observed within a population.

We now define such a family of belief systems, and subsequently present the individual-level belief systems as particular members of this family.

A **population-level asymmetric belief system** over S comprises a pairwise-interaction belief network, in conjunction with a set of directed edge weights $J \in \mathbb{R}^{N \times N}$, spin thresholds $\boldsymbol{h}_0 \in \mathbb{R}^N$, and threshold adjustment coefficients $\boldsymbol{B} \in \mathbb{R}^{N \times K}$. Formally, we describe a belief system over the elements of S as any tuple

$$\mathcal{M} = \langle S, \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h}_0, \boldsymbol{B} \rangle \tag{7.18}$$

with elements satisfying their above defintions.

The elements of $\boldsymbol{J}$ are such that $J_{ij} \neq 0 \iff A_{ij} = 1$. For a direct influence relation $S_i \to S_j$ between $S_i, S_j \in S$, the sign and magnitude of J_{ij} determine the nature and degree of influence from S_i on the behaviour of S_j. If $J_{ij} > 0$, this induces a tendency for S_j to adopt the previous observed state of S_i. Alternatively, if $J_{ij} < 0$, S_j tends to adopt the opposite state.

The probability that S_j^n adopts $(J_{ij} > 0)$ or opposes $(J_{ij} < 0)$ S_i^{n-1} increases with the magnitude of J_{ij}. If S_j is also influenced by other spins, then the degree to which S_i influences the behaviour of S_j is also determined in-part by the relative magnitude of J_{ij} compared to the other interaction effects.

The threshold terms in $\boldsymbol{h}_0$ describe each spin's tendency toward a particular observed state, irrespective of incoming influences from other spins. In reality, individuals have different experiences, interact with different social crowds, and have different states on unmeasured (but conceivably relevant) cognitive axes. As such, we expect a high degree of heterogeneity between individuals' thresholds for any given cognitive axis, as well as commonalities among individuals with similar contexts. The adjustment coefficient matrix $\boldsymbol{B}$ allows for variation in the spin thresholds, defining a set of multiplicative coefficients for each spin, which are used to update an individual's thresholds with respect to a corresponding set of relevant covariates (e.g., age, education).

For an individual with covariate vector $\boldsymbol{x} \in \mathbb{R}^K$, we define the updated threshold vector as

$$\boldsymbol{h}(\boldsymbol{x}) := \boldsymbol{h}_0 + \boldsymbol{B}\boldsymbol{x} \tag{7.19}$$

The population-level asymmetric belief system is such an example of a family of belief systems parameterised by the adjustment coefficients B. The individual **asymmetric belief system** for an individual with covariate vector $\boldsymbol{x} \in \mathbb{R}^K$ is the tuple:

$$\mathcal{M}(\boldsymbol{x}) := \langle S, \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h}(\boldsymbol{x}) \rangle \tag{7.20}$$

For brevity, we will generally leave the individual parameterisation implicit, denoting a belief system as $\mathcal{M}$, and thresholds as $\boldsymbol{h} \in \mathbb{R}^N$. However, we stress that these should nonetheless be considered as constructs belonging to and varying between individuals.

7.1.4 Model dynamics

Let $\mathcal{M} = \langle S, \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$ be a belief system with $|S| = N$ for $N \in \mathbb{N}$. Suppose that at time $t \in \mathbb{N}$, $\boldsymbol{S}^t = \boldsymbol{s}$ for some $\boldsymbol{s} \in \{-1, +1\}^N$. We consider the dynamics of a spin $S_i \in S$ at time $t + 1$.

The behaviour of S_i^{t+1} is influenced by two driving forces: the threshold h_i, and incoming interaction effects J_{ji} for $j \neq i$. Suppose that the net effect of incoming interaction effects is zero, and that $h_i = 0$. In this case S_i^{t+1} is unaffected by other spins and has no inherent tendency toward -1 or $+1$. Naturally, we expect to observe either state with equal probability:

$$\frac{1}{2} = P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}(-1) = P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}(+1) \tag{7.21}$$

Suppose instead that $h_i \neq 0$. Now S_i^{t+1} has an increased tendency toward the state sharing the same sign as h_i:

$$P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}(\operatorname{sgn}(h_i)) > P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}(-\operatorname{sgn}(h_i)) \tag{7.22}$$

For instance, if $h_i > 0$ then $S_i^{t+1} = 1$ with probability greater than $\frac{1}{2}$.

Now suppose that $h_i = 0$, and that the net effect of incoming interaction effects toward S_i is non-zero. Recall that for a given interaction effect J_{ji}, S_i^{t+1} will tend to adopt the previous observed state of S_j^t if $J_{ji} > 0$, and the opposite state otherwise. Thus S_i^{t+1} is influenced to adopt a state with the same sign as $J_{ji} \cdot s_j^t$.

Moreover, recall that the degree to which S_j^t influences S_i^{t+1} increases with $|J_{ji}|$. We can then consider the linear combination of incoming influences with the prior observed state as a net 'effective' influence on S_i^{t+1}:

$$J_i^{\text{eff}}(\boldsymbol{s}) := \sum_{j=1}^{N} A_{ji} J_{ji} s_j \tag{7.23}$$

As before, S_i^{t+1} experiences a tendency to adopt the state which shares the same sign as this term:

$$P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}(\operatorname{sgn}\big(J_i^{\text{eff}}(\boldsymbol{s})\big)) > P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}(-\operatorname{sgn}\big(J_i^{\text{eff}}(\boldsymbol{s})\big)) \tag{7.24}$$

Finally, if h_i is also non-zero, we can straightforwardly include its effect as an additional term. We define the effective threshold on S_i^{t+1} as:

$$h_i^{\text{eff}}(\boldsymbol{s}) := h_i + \sum_{j=1}^{N} A_{ji} J_{ji} s_j \tag{7.25}$$

and have

$$P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}(\operatorname{sgn}\big(h_i^{\text{eff}}(\boldsymbol{s})\big)) > P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}(-\operatorname{sgn}\big(h_i^{\text{eff}}(\boldsymbol{s})\big)) \tag{7.26}$$

Until this point we have left $P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}$ unspecified. As in the typical Ising model formulation, we take the probability as given by the Boltzmann distribution. However, unlike in the standard formulation, we parameterise this distribution with the energy of S_i^{t+1} given the previous configuration, which may be interpreted as the **energy required to transition**. For $s^{t+1} \in \{-1, +1\}$, we define this term as:

$$\begin{aligned} H_i(s^{t+1}|\boldsymbol{s}) &:= -h_i \cdot s^{t+1} - \sum_{j=1}^{N} A_{ji} J_{ji} s_j \cdot s^{t+1} && (7.27.1) \\ &= -s^{t+1} \cdot h_i^{\text{eff}(\boldsymbol{s})} && (7.27.2) \end{aligned}$$

The probability that $S_i^{t+1} = s^{t+1}$ is then given by:

$$P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}(s^{t+1}) := \frac{\exp\left[-\frac{1}{T} \cdot H_i(s^{t+1}|\boldsymbol{s})\right]}{\exp\left[-\frac{1}{T} \cdot H_i(-1|\boldsymbol{s})\right] + \exp\left[-\frac{1}{T} \cdot H_i(+1|\boldsymbol{s})\right]} \tag{7.28}$$

Where the temperature parameter $T \in \mathbb{R}_{>0}$ controls the degree of stochasticity. At high temperatures Equation 7.28 converges to a uniform distribution over $\{-1, +1\}$ (i.e., maximum stochasticity), whereas as $T \to 0^+$ it converges in probability (**TODO: double-check this statement**) to a distribution over the restricted set of states with minimum energy.

Applying Equation 7.27 and substituting $s^{t+1} = 1$, Equation 7.28 reduces to a logistic distribution:

$$P_{S_i^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}}(+1) = \frac{\exp\left[h_i^{\text{eff}}(\boldsymbol{s})/T\right]}{\exp\left[-h_i^{\text{eff}}(\boldsymbol{s})/T\right] + \exp\left[h_i^{\text{eff}}(\boldsymbol{s})/T\right]} \tag{7.29.1}$$

$$= \text{logistic}\left(h_i^{\text{eff}}(\boldsymbol{s})/T\right) \tag{7.29.2}$$

We illustrate the relationship between $h_i^{\text{eff}}(\boldsymbol{s})$ and the the probability that S_i^{t+1} adopts the state $+1$ in Figure 7.6. The observed behaviour matches our expectations per our above discussion. When $h_i^{\text{eff}}(\boldsymbol{s}) = 0$ (corresponding, for instance, to a scenario $h_i = 0$ and net-zero incoming influences) the probability of either state is $1/2$. As the effective threshold increases along the positive axis or decreases along the negative axis, the probability that $S_i^{t+1} = +1$ increases or decreases respectively. Notice also that at higher temperatures a larger-magnitude effective threshold is required to achieve the same probability observed at lower temperatures.

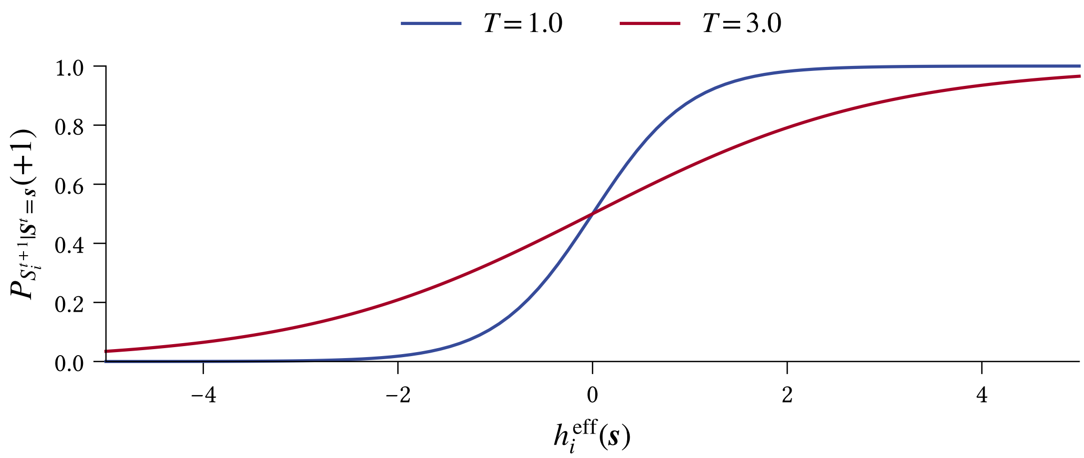

Figure 7.6: The probability that a spin S_i adopts the state $+1$ at time $t+1$ for $t \in \mathbb{N}$ increases with the effective threshold on S_i^{t+1} according to a logistic distribution per Equation 7.29, with greater stochasticity observed at higher temperatures.

The instantaneous observed states for any pair of spins $S_i, S_j \in S$ at time $t+1 \in \mathbb{N}_{>0}$ are conditionally independent random variables given the previous state $\boldsymbol{s}^t$. Thus the full distribution over configurations at time $t+1$ is given by:

$$P_{\boldsymbol{S}^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}^t}\left(\boldsymbol{s}^{t+1}\right) := \prod_{i=1}^{N} \frac{\exp\left[s_i^{t+1} \cdot h_i^{\text{eff}}(\boldsymbol{s}^t)/T\right]}{\exp\left[-h_i^{\text{eff}}(\boldsymbol{s}^t)/T\right] + \exp\left[h_i^{\text{eff}}(\boldsymbol{s}^t)/T\right]} \tag{7.30}$$

This further simplifies as follows (Nguyen et al., 2017):

$$P_{\boldsymbol{S}^{t+1}|\boldsymbol{S}^t=\boldsymbol{s}^t}\left(\boldsymbol{s}^{t+1}\right) = \frac{\exp\left[\sum_{i=1}^{N} s_i^{t+1} \cdot h_i^{\text{eff}}(\boldsymbol{s}^t)/T\right]}{\prod_{i=1}^{N} 2\cosh\left[h_i^{\text{eff}}(\boldsymbol{s}^t)/T\right]} \tag{7.31}$$

7.1.5 Inverse problem

Consider a population of $M \in \mathbb{N}$ individuals, with population-level belief system $\mathcal{M} = \langle S, \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h}_0, \boldsymbol{B} \rangle$ for a known set of cognitive axes S with $|S| = N$ for $N \in \mathbb{N}$, but for which the remaining parameters are unknown.

Suppose that we have observed, for each individual $m \in [1, M]$, a sequence of evenly-spaced configurations, $\left\{ \boldsymbol{s}^{\tau}_{(m)} \right\}_{\tau=1}^{t}$, from the individual belief system $\mathcal{M}(\boldsymbol{x}_m)$, where the vector $\boldsymbol{x}_m \in \mathbb{R}^K$ parameterises $\mathcal{M}$ in accordance with contextual factors specific to m.

If the existence of pairwise relations $\boldsymbol{A}$ is known, we can use maximum likelihood estimation to recover a model $\widetilde{\mathcal{M}} = \langle S, \boldsymbol{A}, \tilde{\boldsymbol{J}}, \tilde{\boldsymbol{h}}_0, \tilde{\boldsymbol{B}} \rangle$ which approximates $\mathcal{M}$. Specifically, we choose $\widetilde{\mathcal{M}}$ such that the probability of the observed data given $\widetilde{\mathcal{M}}$ is maximised:

$$\langle \tilde{\boldsymbol{J}}, \tilde{\boldsymbol{h}}_0, \tilde{\boldsymbol{B}} \rangle = \underset{\boldsymbol{J}', \boldsymbol{h}'_0, \boldsymbol{B}'}{\operatorname{argmax}} P_{\mathcal{M}'} \left(\left\{ \boldsymbol{s}^{\tau}_{(1)} \right\}_{\tau} \ldots \left\{ \boldsymbol{s}^{\tau}_{(m)} \right\}_{\tau} \mid \mathcal{M}' = \langle S, \boldsymbol{A}, \boldsymbol{J}', \boldsymbol{h}'_0, \boldsymbol{B}' \rangle \right) \tag{7.32}$$

Consider a given individual indexed by $m \in [1, M]$. From Equation 7.31 we have that the probability of observing a configuration $\boldsymbol{s}^{t+1}_{(m)}$ at time $t + 1 \in \mathbb{N}_{>1}$ depends only on the previous configuration $\boldsymbol{s}^{t}_{(m)}$. Therefore the probability of observing a sequence of configurations $\left\{ \boldsymbol{s}^{\tau}_{(m)} \right\}_{\tau=1}^{t}$ under a candidate model $\mathcal{M}'$ is simply the product of conditional probabilities between each pair of consecutive timepoints:

$$\begin{aligned} P\left(\left\{ \boldsymbol{S}^{\tau}_{(m)} \right\}_{\tau=1}^{t} = \left\{ \boldsymbol{s}^{\tau}_{(m)} \right\}_{\tau=1}^{t} \right) &= P_{\boldsymbol{S}^{1}_{(m)} \ldots \boldsymbol{S}^{t}_{(m)}} \left(\boldsymbol{s}^{1}_{(m)} \ldots \boldsymbol{s}^{t}_{(m)} \right) && (7.33.1) \\ &= \prod_{\tau=1}^{t-1} P_{\boldsymbol{S}^{\tau+1}_{(m)} \mid \boldsymbol{S}^{1}_{(m)} = \boldsymbol{s}^{1}_{(m)} \ldots \boldsymbol{S}^{\tau}_{(m)} = \boldsymbol{s}^{\tau}_{(m)}} \left(\boldsymbol{s}^{\tau+1}_{(m)} \right) && (7.33.2) \\ &= \prod_{\tau=1}^{t-1} P_{\boldsymbol{S}^{\tau+1}_{(m)} \mid \boldsymbol{S}^{\tau}_{(m)} = \boldsymbol{s}^{\tau}_{(m)}} \left(\boldsymbol{s}^{\tau+1}_{(m)} \right) && (7.33.3) \end{aligned}$$

If individuals are assumed independent, such that the state of any individual does not influence the future states of other individuals, then Equation 7.32 reduces to a product over the likelihoods for each separate individual:

$$L_D(\mathcal{M}') = P_{\mathcal{D}}(D; \mathcal{M}') = \prod_{m=1}^{M} \prod_{\tau=1}^{t-1} P_{\boldsymbol{S}^{\tau+1}_{(m)} \mid \boldsymbol{S}^{\tau}_{(m)} = \boldsymbol{s}^{\tau}_{(m)}} \left(\boldsymbol{s}^{\tau+1}_{(m)} \right) \tag{7.34}$$

In practice, evaluation of Equation 7.34 is often subject to numerical precision errors due to the multiplication of small floating point numbers, thus we instead maximise the *log-likelihood*:

$$\mathcal{L}_D(\mathcal{M}') = \ln(P_{\mathcal{D}}(D;\mathcal{M}')) = \sum_{m=1}^{M}\sum_{\tau=1}^{t-1} \ln P_{\boldsymbol{S}^{\tau+1}_{(m)} \mid \boldsymbol{S}^{\tau}_{(m)}=\boldsymbol{s}^{\tau}_{(m)}}\left(\boldsymbol{s}^{\tau+1}_{(m)}\right) \tag{7.35}$$

Substituting the conditional probability as defined in Equation 7.31, we arrive at a concrete definition for the likelihood:

$$\mathcal{L}_D(\mathcal{M}') = \sum_{m=1}^{M}\sum_{\tau=1}^{t-1}\sum_{i=1}^{N}\left[s^{\tau+1}_{m,i}\cdot h^{\text{eff}}_{m,i}\left(\boldsymbol{s}^{\tau}_{(m)}\right)/T - \ln 2\cosh\left(h^{\text{eff}}_{i}\left(\boldsymbol{s}^{\tau}_{(m)}\right)/T\right)\right] \tag{7.36}$$

We now derive the partial derivatives of $\mathcal{L}_D$ with respect to each parameter considered in the optimisation. Recall that for an individual $m \in [1, M]$ with covariate vector $\boldsymbol{x} \in \mathbb{R}^K$, the effective threshold on a spin $i \in [1, N]$ is defined as:

$$h^{\text{eff}}_{m,i}(\boldsymbol{s},\boldsymbol{x}) = (\boldsymbol{h}_0)_i + \sum_{k=1}^{K} B_{ik}x_k + \sum_{j=1}^{N} A_{ji}J_{ji}s_j \tag{7.37}$$

The partial derivates of the effective threshold with respect to each parameter considered in the optimisation are as follows:

$$\frac{\partial h^{\text{eff}}_i}{\partial(\boldsymbol{h}_0)_\alpha} = \delta_{\alpha,i}, \quad \frac{\partial h^{\text{eff}}_i}{\partial B_{\alpha\beta}} = \delta_{\alpha,i}\left(x_\beta\right), \quad \text{and} \quad \frac{\partial h^{\text{eff}}_i}{\partial J_{\alpha\beta}} = \delta_{\beta,i}(A_{\alpha i}s_\alpha) \tag{7.38}$$

where $\delta_{i,j}$ denotes the kronecker delta.

Let p be an arbitrary parameter in $\boldsymbol{J}'$, $\boldsymbol{h}'_0$, or $\boldsymbol{B}'$, then the partial derivative of $\mathcal{L}_D$ with respect to p takes the form:

$$\frac{\partial\mathcal{L}_D}{\partial p} = \frac{1}{T}\sum_{m,\tau,i}^{M,t-1,N}\left(s^{\tau+1}_{m,i} - \tanh\left(h^{\text{eff}}_{m,i}(\boldsymbol{s}^{\tau}_{m})/T\right)\right)\cdot\frac{\partial}{\partial p}h^{\text{eff}}_{m,i}(\boldsymbol{s}^{\tau}_{m}) \tag{7.39}$$

Substituting the corresponding partial derivatives from Equation 7.38, we find that the partial derivatives of the likelihood with respect to each parameter type are as follows:

$$\frac{\partial\mathcal{L}_D}{\partial(\boldsymbol{h}_0)_i} = \frac{1}{T}\sum_{m,\tau} s^{\tau+1}_{m,i} - \tanh\left(h^{\text{eff}}_{m,i}(\boldsymbol{s}^{\tau}_{m})/T\right) \tag{7.40.1}$$

$$\frac{\partial\mathcal{L}_D}{\partial B_{ik}} = \frac{1}{T}\sum_{m,\tau}\left(s^{\tau+1}_{m,i} - \tanh\left(h^{\text{eff}}_{m,i}(\boldsymbol{s}^{\tau}_{m})/T\right)\right)\cdot x^{(m)}_k \tag{7.40.2}$$

$$\frac{\partial\mathcal{L}_D}{\partial J_{ij}} = \frac{1}{T}\sum_{m,\tau}\left(s^{\tau+1}_{m,j} - \tanh\left(h^{\text{eff}}_{m,j}(\boldsymbol{s}^{\tau}_{m})/T\right)\right)\cdot A_{ij}s^{\tau}_{m,i} \tag{7.40.3}$$

The likelihood is maximised when $\frac{\partial \mathcal{L}_D}{\partial (\boldsymbol{h}_0)_i} = \frac{\partial \mathcal{L}_D}{\partial J_{i,j}} = \frac{\partial \mathcal{L}_D}{\partial B_{i,k}} = 0$ for all $i, j \in [1, N]$ and $k \in [1, K]$.

If the relational structure $\boldsymbol{A}$ is unknown, we can either assume a fully-connected structure, such that for each pair of cognitive axes $S_i, S_j \in S$ we have $S_i \rightarrow S_j$, or infer the relational structure, for instance using causal discovery methods.

7.1.6 Simulating interventions

Interventions on a belief system $\mathcal{M} = \langle S, \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$ can manifest as either influence toward a particular state for a given belief or attitude, or as a 'rewiring' of the causal interactions between cognitive axes.

The first category of intervention, which we refer to as a **threshold intervention**, is modelled as an adjustment to the threshold vector $\boldsymbol{h}$. For a spin $S_i \in S$, we model a threshold intervention toward the state +1 as the adjusted model $\mathcal{M}' = \langle S, \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h}' \rangle$, where for $j \in [1, N]$ and an intervention level δ_i:

$$h'_j = \begin{cases} h_j + \delta_i & \text{if } j = i \\ h_j & \text{otherwise} \end{cases} \tag{7.41}$$

The second category, which we refer to as a **structural intervention**, is modelled as an adjustment to the relational structure or causal influences between spins. For a pair of spins $S_i, S_j \in S$ and an intervention level δ_{ij}, a structural intervention is modelled as the adjusted model $\mathcal{M}' = \langle S, \boldsymbol{A}, \boldsymbol{J}', \boldsymbol{h} \rangle$ where

$$J'_{k\ell} = \begin{cases} J_{ij} + \delta_{ij} & \text{if } (k, \ell) = (i, j) \\ J_{k\ell} & \text{otherwise} \end{cases} \tag{7.42}$$

7.2 Asymmetric belief system model

We consider a belief system as comprises a collection of **cognitive axes**, arranged in a directed network, with edges describing pairwise causal relations. Since the directed edges are permitted to vary in effect strength, standard methods for solving the inverse Ising problem at equilibrium at not applicable here. Instead we adopt a maximum-likelihood parameter reconstruction method using time-series data to identify a maximum-entropy model which matches the (pairwise) time-lagged correlations observed in the data.

7.2.1 Cognitive axes

Cognitive axes are either:

- **Beliefs:** Epistemic positions regarding states of affairs (e.g., 'climate change is real' or 'extreme weather events are becoming more frequent'), or
- **Attitudes:** Evaluative positions (e.g., policy support or opposition or emotive states such as 'happy' or 'sad').

As implied by the name, for the purposes of this study we consider cognitive *axes* with polarity, i.e., with contrasting possible states representing opposite ends of a spectrum. For instance, consider the epistemic position:

Believes that climate change is happening.

If we consider the 'opposite' position, there are at least two reasonable choices:

Does not *believe that climate change is happening.*

and

Believes that climate change is ***not*** *happening.*

Let $\mathscr{B}(p)$ denote the belief that the predicate p is true. For simplicity, we will assume here that $\mathscr{B}(p)$ and $\mathscr{B}(\neg p)$ are mutually exclusive, such that an individual cannot simultaneously hold two conflicting epistemic positions[3]. The first choice is then equivalently stated as $\neg\mathscr{B}(p)$, and the second choice as $\mathscr{B}(\neg p)$, where $p :=$ climate change is happening.

Observe that in the first choice, $\neg\mathscr{B}(p)$, the action of believing *per se* is negated. Thus the statement is satisfied so long as the individual does not hold the belief p. Due to the

[3]There is evidence to suggest that in reality individuals do often hold conflicting epistemic positions, with only one being 'active' at any given point in time, depending on contextual factors.

mutual exclusivity of $\mathscr{b}(p)$ and $\mathscr{b}(\neg p)$, this is of course realisable when the individual holds the belief $\mathscr{b}(\neg p)$; however, it is also realised when the individual holds *no* belief on this statement whatsoever. Hence the first choice does not have the desired polarity property, since it allows for the absence of belief (or analogously, an ambivalent attitude).

In the second choice, since the belief itself is not negated, we require at least one of $\mathscr{b}(p)$ or $\mathscr{b}(\neg p)$ to be true. By the mutual exclusivity of these options, it follows that the second choice has our desired polarity property.

7.2.2 Belief system interactions

Directed edges describe the direct causal relations between cognitive axes. These causal relations operate across timesteps, such that the instantaneous state of a belief system influences the subsequent state. We formally define a direct causal relation $a \to b$ as a conditional dependence in the distribution over possible states of b on the previous value of a.

Definition 7.2.1 (Direct causal relation)

Let $\mathcal{M}$ be a pairwise belief system, and $s_i, s_j \in S$ be a pair of cognitive items. We say there is a **direct causal relation** $s_i \to s_j$ if, and only if, given a configuration $\boldsymbol{s}^t$, for some possible value s of s_j,

$$\mathbb{P}\left[s_j^{t+\mathrm{d}t} = s \mid \boldsymbol{s}^t\right] \neq \mathbb{P}\left[s_j^{t+\mathrm{d}t} = s \mid \boldsymbol{s}_{-i}^t\right] \tag{7.43}$$

♣

Consider an $a \to b$ between cognitive axes a and b, with weight $w_{a\to b}$. The sign of $w_{a\to b}$ describes the tendency for b to adopt the same (+) or opposite (−) polarity that a exhibited in the previous timestep. The magnitude $|w_{a\to b}|$ describes the degree to which this relation influences the behaviour of b. Even if $|w_{a\to b}|$ is relatively large in absolute terms, if it is small compared to another relation $|w_{c\to b}|$, then the influence of a on b may be limited.

Unlike in the equilibrium (symmetric) Ising model, two spins may be uncorrelated within a given timestep, even though their behaviour is highly correlated in time. For example, consider the model shown in Figure 7.7, comprising two cognitive axes, s_A and s_B. The behaviour of s_A is entirely unconstrained by s_B, yet the large magnitude relation $s_A \to s_B$ ensures the behaviour of s_B is almost entirely set by s_A. In particular, at each timestep, s_B adopts the previous state of s_A with high probability.

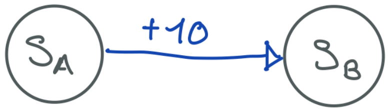

Figure 7.7: A two-spin asymmetric belief system in which the instantaneous correlation between spins is zero, but the time-delayed correlation is non-zero.

Let $\{\boldsymbol{s}\}_{t=1}^{m}$ be a sequence of samples drawn from this model, with m sufficiently large, and define $\{s_A\}_{t=1}^{m}$ and $\{s_B\}_{t=1}^{m}$ as the sequences of s_A and s_B components respectively. Since the behaviour of s_A is unconstrained, we find that the instantaneous correlation between s_A and s_B is zero:

$$\mathrm{Corr}\Big(\{s_A\}_{t=1}^{m}, \{s_B\}_{t=1}^{m}\Big) = 0 \tag{7.44}$$

yet the time-delayed correlation between s_B and the *previous state* of s_A is close to its maximum possible value:

$$\mathrm{Corr}\Big(\{s_A\}_{t=1}^{m-1}, \{s_B\}_{t=2}^{m}\Big) \approx 1 \tag{7.45}$$

We permit self-interaction relations which operate in the same fashion, representing the intertia of a cognitive axis. Negative weights for self-interaction relations reflect an inherent tendency for a cognitive axis to fluctuate. As we have defined cognitive axes as having polarity, such fluctuations would reflect dramatic shifts in beliefs or attitudes, which we do not expect to observe in reality. Thus we expect self-interaction weights to be non-negative in general. A large positive self-interaction weight describes a tendency for the corresponding cognitive axis to retain its previously-measured state, even under pressure from other cognitive axes. This captures the notion that some beliefs or attitudes have inherently slower timescales than others, e.g., an individual's political ideology may be expected to vary more slowly than their attitudes toward particular policies or political candidates.

An alternative interpretation of the causal relations between cognitive axes is as conditional non-independence relations.

Conjecture 7.2.2

Let $\mathcal{M}$ be a pairwise belief system with cognitive axes S, and suppose that all configurations of the belief system state occur with non-zero probability. For any pair of cognitive axes $s_i, s_j \in S$, there exists a directed causal relation $s_i \to s_j$ if, and only if, $s_j^{t+\mathrm{d}t}$ and s_i^t are conditionally non-independent. ♡

Proof. Let $\boldsymbol{s}^t$ be the instantaneous configuration of cognitive axes in the belief system at time t. Applying Bayes' rule to the conditional probability distribution over $s_j^{t+\mathrm{d}t}$ given the previous system configuration, we find

$$\begin{aligned}
\mathbb{P}\left[s_j^{t+\mathrm{d}t} \mid \boldsymbol{s}^t\right] &= \frac{\mathbb{P}\left[s^{t+\mathrm{d}t}, \boldsymbol{s}^t\right]}{\mathbb{P}\left[\boldsymbol{s}^t\right]} && \text{Bayes rule} \\
&= \frac{\mathbb{P}\left[s_j^{t+\mathrm{d}t}, s_i^t \mid \boldsymbol{s}_{-i}^t\right]\mathbb{P}\left[\boldsymbol{s}_{-i}^t\right]}{\mathbb{P}\left[s_i^t \mid \boldsymbol{s}_{-i}^t\right]\mathbb{P}\left[\boldsymbol{s}_{-i}^t\right]} && \text{Bayes rule (again)} \\
&= \frac{\mathbb{P}\left[s_j^{t+\mathrm{d}t}, s_i^t \mid \boldsymbol{s}_{-i}^t\right]}{\mathbb{P}\left[s_i^t \mid \boldsymbol{s}_{-i}^t\right]} && 0 < \mathbb{P}\left[\boldsymbol{s}_i^t\right]
\end{aligned}$$

Combining this result with Definition 7.2.1, it follows that s_i *does not* directly influence s_j if, and only if,

$$\mathbb{P}\left[s_j^{t+\mathrm{d}t}, s_i^t \mid \boldsymbol{s}_{-i}^t\right] = \mathbb{P}\left[s_i^t \mid \boldsymbol{s}_{-i}^t\right] \cdot \mathbb{P}\left[s_j^{t+\mathrm{d}t} \mid \boldsymbol{s}^t\right]$$

i.e., when $s_j^{t+\mathrm{d}t}$ is independent of s_i^t conditional on $\boldsymbol{s}_{-i}^t$. The main result then follows directly by taking the converse. □

7.2.3 Belief system dynamics

7.2.3.1 Model components

Let S be a set of $n \in \mathbb{N}$ cognitive axes, arranged as nodes on a directed network with adjacency matrix $\boldsymbol{A} \in \{0,1\}^{n\times n}$ and edge weights $\boldsymbol{J} \in \mathbb{R}^{n\times n}$. At time t, each node (spin) $S_i \in S$ adopts a state $S_i^t = s$ for $s \in \{-1, +1\}$.

For a pair of nodes (spins) $s_i, s_j \in S$, the existence of an edge $s_i \to s_j$ reflects a directed causal relationship, such that the instantaneous state of S_i influences the subsequent state of S_j. The sign of J_{ij} determines the nature of this influence. In particular, $S_j^{t+1} = \mathrm{sign}(J_{ij}) \cdot S_i^t$ with increased probability. The degree of increase in probability increased with $|J_{ij}|$, both in absolute terms, and relative to the weights of other edges into S_j.

For each spin we define a **threshold** which captures unmeasured influences on individuals' tendences toward particular states $s \in \{-1, +1\}$. In reality, individuals have different experiences, interact with different social crowds, and have different states on unmeasured (but conceivably relevant) cognitive axes. As such, we would expect a high degree of heterogeneity between individuals in the threshold for any given cognitive axis. On the other hand, we expect to observe commonalities between individuals from similar contexts. When per-individual observational data is limited, allowing thresholds to vary as a function of individual context provides a reasonable middle-ground between ignoring individual differences and fitting per-individual thresholds.

To model the thresholds, we define a vector of baseline effects $\boldsymbol{h}_0 \in \mathbb{R}^n$ and a matrix of contextual adjustment coefficients $\boldsymbol{B} \in \mathbb{R}^{n \times k}$, where $k \in \mathbb{N}$ is the number of covariates parameterising the thresholds. For an individual with covariates $\boldsymbol{x} \in \mathbb{R}^k$, we then define their threshold vector as:

$$\boldsymbol{h}(\boldsymbol{x}) \coloneqq \boldsymbol{h}_0 + \boldsymbol{B}x \tag{7.46}$$

For clarity, we will often omit explicit mention of the covariate vector, instead referring directly to $\boldsymbol{h}$ as a vector of thresholds.

7.2.3.2 System transitions

The probability of a spin $S_i \in S$ adopting a particular state $s \in \{-1, +1\}$ is set by the energy differential with the alternative state, where lower-energy states are preferred. We define the local energy experienced by S_i when adopting the value s, given the previous belief system configuration, as:

$$H\left(S_i^{t+\mathrm{d}t} = s | \boldsymbol{s}^t\right) = -h_i s - \sum_{j=1}^{n} A_{ji} J_{ji} s_j s \tag{7.47.1}$$

$$= -s \cdot h_i^{\mathrm{eff}}(\boldsymbol{s}^t) \tag{7.47.2}$$

Where $h_i^{\mathrm{eff}}(\boldsymbol{s}^t)$ is the effective local field imposed on s_i at time t, comprising the accumulation of effects which result in an overall tendency toward $s_i^{t+\mathrm{d}t} \in \{-1, +1\}$, and is defined as

$$h_i^{\mathrm{eff}}(\boldsymbol{s}^t) = h_i + \sum_{j=1}^{n} A_{ji} J_{ji} s_j \tag{7.48}$$

The dynamics of S_i are succinctly described through a conditional probability distribution given the previous system state (Nguyen et al., 2017)

$$\mathbb{P}\Big[S_i^{t+\mathrm{d}t} = s \mid \boldsymbol{S}^t = \boldsymbol{s}\Big] = \frac{\exp\left[-\frac{1}{T} \cdot s \cdot h_i^{\mathrm{eff}}(\boldsymbol{s})\right]}{\exp\left[\frac{1}{T} \cdot h_i^{\mathrm{eff}}(\boldsymbol{s})\right] + \exp\left[-\frac{1}{T} \cdot h_i^{\mathrm{eff}}(\boldsymbol{s})\right]} \tag{7.49}$$

Where the temperature parameter $T \in \mathbb{R}_{>0}$ controls the degree of stochasticity. At high temperatures Equation 7.49 converges to a uniform distribution over $\{-1, +1\}$ (i.e., maximum stochasticity), whereas as $T \to 0^+$ it converges in probability (**TODO: double-check this statement**) to a distribution over the restricted set of states with minimum energy.

At each timestep, we allow *every* spin the opportunity to update. This is known as **synchronous updating**. The alternative is **asynchronous updating**, in which a singular spin is randomly sampled to update on each timestep. Asynchronous updates are typically preferrered, both for model realism in case the real phenomenon exhibits continuous-time updates, and to avoid degenerate behaviours such as all spins fluctuating between two system configurations. We use synchronous updates here for two reasons. Firstly, the likelihood of such degenerate behaviour is unlikely due to the inclusion of self-interaction effects, which provide inherent, heterogeneous timescales to the modelled cognitive axes. Secondly, assuming synchronicity significantly simplifies the inverse problem, as will be discussed in Chapter 7.2.4.

Observe that for $s_i \neq s_j \in S$, the respective conditional distributions over $S_i^{t+\mathrm{d}t}$ and $S_j^{t+\mathrm{d}t}$ depend only on the *previous* timestep, so are conditionally independent given $\boldsymbol{S}^t = \boldsymbol{s}$. Thus, given the synchronous update assumption, we can straightforwardly extend Equation 7.49 to a conditional distribution over the entire belief system configuration given the previous configuration:

$$\mathbb{P}\big[\boldsymbol{S}^{t+\mathrm{d}t} = \boldsymbol{s}' \mid \boldsymbol{S}^t = \boldsymbol{s}\big] = \prod_{i=1}^{n} \mathbb{P}\Big[S_i^{t+\mathrm{d}t} = s_i' \mid \boldsymbol{S}^t = \boldsymbol{s}\Big] \tag{7.50.1}$$

$$= \prod_{i=1}^{n} \frac{\exp\left[-\frac{1}{T} \cdot s_i' \cdot h_i^{\mathrm{eff}}(\boldsymbol{s})\right]}{\exp\left[\frac{1}{T} \cdot h_i^{\mathrm{eff}}(\boldsymbol{s})\right] + \exp\left[-\frac{1}{T} \cdot h_i^{\mathrm{eff}}(\boldsymbol{s})\right]} \tag{7.50.2}$$

Furthermore, recognising Equation 7.49 as a logistic function with codomain $\{-1, +1\}$, we can write for each spin $s_i \in S$:

$$\frac{1}{2}\Big(S_i^{t+\mathrm{d}t} + 1\Big) \sim \mathrm{logistic}\left[-\frac{1}{T} \cdot h_i^{\mathrm{eff}}(\boldsymbol{s}^t)\right] \tag{7.51}$$

7.2.4 Inverse problem

Consider a belief system comprising $n \in \mathbb{N}$ cognitive axes $\boldsymbol{S}$, and a dataset $\boldsymbol{D}$ with observations from $k \in \mathbb{N}$ individuals, with each row containing measurements for each of the n cognitive axes. The inverse problem consists in identifying parameters $\boldsymbol{A} \in \{0,1\}^{n\times n}, \boldsymbol{J} \in \mathbb{R}^{n\times n}, \vec{h} \in \mathbb{R}^n$ such that the resulting model is a plausible candidate for the mechanism which generated $\boldsymbol{D}$. In other words, the inverse problem is the task of inferring the belief system which generated the observed data.

This process consists in two parts: (i) determining the belief system relational structure $\boldsymbol{A}$, and (ii) estimating the effect sizes $\boldsymbol{J}$ and $\vec{h}$.

To infer the effect sizes for interactions and thresholds, we use maximum likelihood estimation, choosing a model parameterisation which maximises the probability of observing the dataset D. Maximum likelihood estimation using cross-sectional observational data is commonly used to solve the inverse problem for the symmetric (equilibrium) Ising model.

In the case of the asymmetric belief system model described above, the model dynamics are defined in terms of a conditional probability distribution over system states, given the previous state. As such, we instead need to maximise the conditional likelihood:

$$\mathcal{L}_{\boldsymbol{S}^{t+1} \mid \boldsymbol{S}^t}\left(\boldsymbol{A}, \boldsymbol{J}, \vec{h}\right) := \frac{1}{km} \sum_{r=1}^{k} \sum_{\tau=1}^{m-1} \ln \mathbb{P}\left[\boldsymbol{S}_r^{\tau+1} = D_{r,\tau+1} \mid \boldsymbol{S}_r^{\tau} = D_{r,\tau}\right] \tag{7.52}$$

However, cross-sectional data is not sufficient for this task, as computing (let alone maximising) the conditional likelihood requires repeated observations from each individual. Thus to solve the inverse Ising problem for the asymmetric belief system model, we require time-series observational data.

Note: There are a couple of other things to discuss here, which I am still working on. Notably:

1. That time-series observations are required to infer self-interaction effects, and
2. That attempting to do non-conditional maximum likelihood estimation also fails with cross-sectional data, because the problem is underdetermined (we have more unknowns than observables).

7.3 Belief system interactions

We now formalise the notion of a pairwise-interaction belief system. Consider a finite set of cognitive items, S, which may include a mixture of belief and attitude axes, and let S form the set of vertices in a directed *pairwise belief system network* $G_{\mathcal{M}} = \langle S, E \rangle$. For each ordered pair of vertices $s, s' \in S$, there exists an edge $(s, s') \in E$ if, and only if, the state of the cognitive item s' is subject to **direct influence** from s.

This notion of direct influence is presently ambiguous, and so it is worthwhile to clarify exactly what is meant here. We say that s **influences** s' if the instantaneous state of s affects the subsequent state distribution of s'. For instance, if I expect that it will rain this afternoon, then my attitude toward bringing an umbrella to work is positive. On the other hand, if I expect fine weather, then bringing an umbrella to work is an unnecessary nuisance. Hence my attitude toward the umbrella is influenced by my expectations about the weather. If I live relatively close to the office, then my attitude toward commuting on public transport may be similarly influenced by these expectations; however, if I live far away, such that biking to work isn't feasible, then this influence relation may no longer obtain.

An influence relation is said to be **direct** if it persists after conditioning on all other relevant factors. If I expect poor weather this afternoon, and consequently consider taking my umbrella to work, I may conceivably worry about leaving my umbrella on the bus. My expectations about the weather therefore influence my concern about losing my umbrella, but this influence is indirect; after conditioning on my attitude toward taking my umbrella to work, my concern is independent of my expectations about the weather. We formalise this notion in Definition 7.2.1.

Under this interpretation it is straightforward to show that, if we assume all belief system configurations occur with non-zero probability, then the direct influence relation $\sigma_i \to \sigma_j$ is equivalent to $\sigma_j^{t+\delta t}$ being conditionally non-independent of σ_i^t, given all other cognitive items at time t.

Conjecture 7.3.1

Let $G_{\mathcal{M}} = \langle S, E \rangle$ be a pairwise belief system network, and suppose that all belief system state configurations have non-zero probability of occurring. For any pair of cognitive items $\sigma_i \neq \sigma_j \in S$, σ_i directly influences σ_j if, and only if, $\sigma_j^{t+\delta t} \not\perp\!\!\!\perp \sigma_i^t \mid \boldsymbol{\sigma}_{-i}^t$

♡

Proof. Let $\boldsymbol{\sigma}^t$ be the instantaneous configuration of cognitive item states in the belief system network at time t, then applying Bayes' rule to the conditional probability distribution over $\sigma_j^{t+\delta t}$ given the previous system configuration, we find

$$\begin{aligned}
\mathbb{P}\left[\sigma_j^{t+\delta t} \mid \boldsymbol{\sigma}^t\right] &= \frac{\mathbb{P}\left[\sigma_j^{t+\delta t}, \boldsymbol{\sigma}^t\right]}{\mathbb{P}[\boldsymbol{\sigma}^t]} && \text{Bayes rule} \\
&= \frac{\mathbb{P}\left[\sigma_j^{t+\delta t}, \sigma_i^t \mid \boldsymbol{\sigma}_{-i}^t\right]\mathbb{P}[\boldsymbol{\sigma}_{-i}^t]}{\mathbb{P}[\sigma_i^t \mid \boldsymbol{\sigma}_{-i}^t]\mathbb{P}[\boldsymbol{\sigma}_{-i}^t]} && \text{Bayes rule (again)} \\
&= \frac{\mathbb{P}\left[\sigma_j^{t+\delta t}, \sigma_i^t \mid \boldsymbol{\sigma}_{-i}^t\right]}{\mathbb{P}[\sigma_i^t \mid \boldsymbol{\sigma}_{-i}^t]} && 0 < \mathbb{P}[\boldsymbol{\sigma}_{\boldsymbol{i}}^{\boldsymbol{t}}]
\end{aligned}$$

Combining this result with Definition 7.2.1, it follows that σ_i *does not* directly influence σ_j if, and only if,

$$\mathbb{P}\left[\sigma_j^{t+\delta t}, \sigma_i^t \mid \boldsymbol{\sigma}_{-i}^t\right] = \mathbb{P}[\sigma_i^t \mid \boldsymbol{\sigma}_{-i}^t] \cdot \mathbb{P}\left[\sigma_j^{t+\delta t} \mid \boldsymbol{\sigma}_{-i}^t\right]$$

i.e., when $\sigma_j^{t+\delta t}$ is independent of σ_i^t conditional on $\boldsymbol{\sigma}_{-i}^t$. The main result then follows directly by taking the converse. □

We associate a signed weight with each direct influence relation, with the sign specifying whether the head tends to assume the same value (+) as the tail or the opposite value (−), and the magnitude describing the degree to which a relation influences the behaviour of the head. Finally, we include a signed baseline effect for each cognitive item, which describes the tendency for that item to assume a positive or negative state in the case that the combined incoming direct influence relations have a net-zero influence. With the conceptual foundations now laid, we present our formal pairwise-interaction belief system model in Definition 7.3.2.

Definition 7.3.2 (Pairwise-interaction belief system)

A **pairwise-interaction belief system** of size $N \in \mathbb{N}$ is any tuple $\mathcal{M} = \langle S, \boldsymbol{A}, \boldsymbol{J}, \vec{h} \rangle$, where

- S is a set of N cognitive items,
- $\boldsymbol{A} \in \{0,1\}^{N\times N}$ specifies the direct influence relations among elements of S as an adjacency matrix,
- $\boldsymbol{J} \in \mathbb{R}^{N\times N}$ contains the weights of the influence relations, and
- $\vec{h} \in \mathbb{R}^N$ is a vector of baseline effects for the elements in S.

♣

7.4 Reviewing belief system models

- Early work, triadic consistency
- Bayesian networks
- Causal attitude network, Ising-style models
- Beliefs as edges vs. beliefs as nodes
- Social influence: hierarchical Ising models, network of belief model
- Within-person vs between-person correlations; inverse problem problems

7.4.1 Symmetric (equillibrium) Ising model

$$P(\boldsymbol{S} = \boldsymbol{s}) = \frac{\exp\left(-\frac{1}{T} H(\boldsymbol{s})\right)}{\sum_{\boldsymbol{s}'} \exp\left(-\frac{1}{T} H(\boldsymbol{s}')\right)} \tag{7.53}$$

Where the temperature parameter $T \in \mathbb{R}_{>0}$ controls the degree of stochasticity. At high temperatures $P(\boldsymbol{S} = \boldsymbol{s})$ converges to a uniform distribution over states (i.e., maximum stochasticity), whereas as $T \to 0^+$ it converges in probability (**TODO: double-check this statement**) to a distribution over the restricted set of states with minimum energy.

- Introduce maximum likelihood parameter estimation using:
  - Cross-sectional methods
  - Time-series methods

7.4.2 Causal Attitude Networks

TODO: Discuss what they say in original paper about differences in the sets of nodes included in different individuals' networks.

The Causal Attitude Network (**CAN**) model (Dalege et al., 2016) is an Ising-style theory of endogenous belief system dynamics, which operates under the assumption that these dynamics are primarily driven by efforts — conscious or otherwise — to reduce cognitive dissonance. The model considers a collection of evaluative axes for cognitive items, such as attitudes, beliefs, feelings, or behaviours, which are arranged as vertices on an undirected, signed, weighted network.

Edges in the network describe causal influences between items, such that a pair of cognitive items related via a positive edge are likely to exhibit evaluations of the same *valence* (positive or negative), while items related via a negative edge are likely to exhibit evaluations with different valences. The degree to which a pair of related items are expected to covary increases with both the absolute magnitude of the weight of their relation, and the relative magnitude compared to other relations involving either of the items. Observations from a pair of items may have low mutual information despite a high absolute magnitude relation, if, for instance, the associated cognitive items have much higher-weight relations with other, non-overlapping sets of cognitive items.

Each cognitive item has an associated **threshold**, $\tau_i \in \mathbb{R}$, describing its tendency to assume either positive ($\tau_i > 0$) or negative ($\tau_i < 0$) evaluations in the absence of influencing interactions with other items, or in the case that the net effect of these interactions is zero.

The CAN model defines the dynamics of a belief system using an equilibrium Ising model framework, with the Hamiltonian of a particular configuration of evaluations defined in Equation 7.54, where $\omega_{ij} \in \mathbb{R}$ is the signed weight of a relation between nodes i and j in the network and $N \in \mathbb{N}$ is the number of cognitive items in the model.

$$H(\boldsymbol{s}) = -\sum_{i=1}^{N} \tau_i s_i - \sum_{\langle ij \rangle} \omega_{ij} s_i s_j \tag{7.54}$$

At equilibrium, the probability is described by the Boltzmann distribution (Equation 7.53) with Equation 7.54 as its Hamiltonian. Since the CAN model is equivalent to a symmetric network Ising model, all standard methods for solving the inverse Ising problem for this class of models also apply here to recovering parameter values $\tau_i, \omega_{ij} \in \mathbb{R}$ for a CAN model assumed to generate the observations in a binary dataset.

7.5 Asymmetric belief systems

We now introduce our belief system model as an extension on the Causal Attitude Network model (Chapter 7.4.2). Our model diverges from this theory in two key aspects: we consider (i) belief systems that are not at equilibrium, and in which (ii) the pairwise interactions between cognitive items may be non-reciprocal, or have different degrees of influence.

Definition 7.5.1 (Asymmetric Causal Attitude Network)

A (pairwise) **asymmetric causal attitude network** (**ACAN**) of size $N \in \mathbb{N}$ is any tuple $\mathcal{M} = \langle S, \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$, where

- S is a set of N cognitive items,
- $\boldsymbol{A} \in \{0,1\}^{N \times N}$ specfies direct influence relations among the elements of S as a directed adjacency matrix,
- $\boldsymbol{J} \in \mathbb{R}^{N \times N}$ contains the weights of the influence relations, with $J_{ij} = 0$ iff. $A_{ij} = 0$, and
- $\boldsymbol{h} \in \mathbb{R}^{N}$ is a vector of threshold effects for the cognitive items in S. ♣

For $N \in \mathbb{N}$, $\mathcal{M} = \langle S, \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$ models a belief system over the cognitive items in S, whose temporal dynamics are conditionally defined with respect to the previous state configuration of cognitive items. Each cognitive item updates probabilistically with transition probabilities in accordance with the resulting relative change in energy. The change in energy experienced by a cognitive item $i \in S$ transitioning from s_i^t to s, for $s_i^t, s \in \{-1,+1\}$, is a combination of the threshold τ_i and incoming interaction effects:

$$H_i(s \mid \boldsymbol{s}^t) = -\tau_i s - \sum_{j=1}^{N} (A_{ji} J_{ji}) s_j^t s \tag{7.55}$$

The common factor s allows us to rewrite this equivalently as a product of s with a term not depending on s. We may consider this latter term as an **effective local field** acting on the cognitive item i at time t:

$$H_i(s \mid \boldsymbol{s}^t) = -s \cdot h_i^{\text{eff}}(\boldsymbol{s}^t) \tag{7.56}$$

where $h_i^{\text{eff}}(\boldsymbol{s}^t) = \tau_i + \sum_{j=1}^{N} A_{ji} J_{ji} s_j^t$. That is to say that we may instead consider the simpler, equivalent scenario in which interaction effects on i from other cognitive items are replaced by a single locally-acting field term.

Derivation?

The probability distribution over possible next states for the cognitive item i is then given by:

$$\mathbb{P}\Big[S_i^{t+\mathrm{d}t} = s \mid \boldsymbol{S}^t = \boldsymbol{s}\Big] = \frac{\exp\big[-\frac{1}{T} \cdot s \cdot h_i^{\text{eff}}(\boldsymbol{s})\big]}{\exp\big[-\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})\big] + \exp\big[\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})\big]} \tag{7.57}$$

where $T \in \mathbb{R}_{>0}$ is a temperature parameter as in Equation 7.53. The symmetry of possible values for $s \in \{-1, +1\}$ allows us to further simplify this expression as follows:

$$\mathbb{P}\Big[S_i^{t+\mathrm{d}t} = s \mid \boldsymbol{S}^t = \boldsymbol{s}\Big] = \frac{\exp\big[-\frac{1}{T} \cdot s \cdot h_i^{\text{eff}}(\boldsymbol{s})\big]}{2\cosh\big[\frac{1}{T} h_i^{\text{eff}}(\boldsymbol{s})\big]} \tag{7.58}$$

We illustrate the behaviour of this probability distribution for a cognitive item i, for varying effective local field strengths and temperature parameters in Figure 7.6. Observe that for high-magnitude negative effective local fields — corresponding to a large positive threshold h_i, and/or strong influences toward positive values from other spins ($J_{ji} s_j > 0$) — the probability that i transitions to $S_i^{t+\mathrm{d}t} = 1$ converges to 1. The opposite effect is observed for high-magnitude positive effective local fields.

Additionally, we clearly see the effect of the temperature parameter on the distribution. In the low-temperature scenario ($T = 1.0$), for effective local field magnitudes $|h_i^{\text{eff}}(s)| \gtrapprox 1.5$, the distribution over possible next states for i is close to constant. As $T \to 0^+$, the curve approches a step function with a threshold at 0. On the other hand, in the high-temperature scenario ($T = 3.0$), the probability of a transition to $S_i^{t+\mathrm{d}t} = -1$ is non-trivial even for high-magnitude negative values of $h_i^{\text{eff}}(\boldsymbol{s})$. For a fixed effective local field, higher temperatures incur transition probability closer to 0.5 (**can include derivation showing entropy $\to$ 1**). Since in any finite model each cognitive item has a bounded effective local field strength, it follows that as $T \to \infty$ the transition probability is decreasingly influenced by the effective local field.

Motivate the differences between this model and the CAN model

Unlike the CAN model, our asymmetric belief system model does not assume equilibrium dynamics. Equation 7.58 defines transition probabilities for a given cognitive item as conditional on the previous system state, with no assumption that this state was at equilibrium, nor at a steady state. As such this formulation is suitable for studying out-of-equilibrium dynamics such as those occurring during interventions.

Let us briefly discuss two aspects of the ACAN model which differ functionally from the symmetric variant. Firstly, we permit interactions between a pair of cognitive items to vary directionally. It may be the case that the relation $A \to B$ exists, yet $B \to A$ does not, or that both exist but with different strengths. How then should we interpret interaction effects in the asymmetric model?

In the symmetric formulation,

Discuss how the dynamics differ

The CAN model, as outlined in (Dalege et al., 2016), is conceptualised as a symmetric (network) Ising model, characterised by its behaviour at equilibrium. In the present study we are interested in questions regarding intervention, in which the goal is to shift an individual's configuraton of beliefs and attitudes such that a particular 'target' cognitive item assumes a desired value, e.g., belief in climate change. These dynamics are inherently non-equilibrium. In the case that an individual's belief system configuration is in an equilibrium steady state, our goal is to disrupt this. If instead their belief system configuration is *not* at equilibrium, then our goal is to influence its dynamics toward our desired steady state. In either case we are fundamentally concerned with the dynamics of belief systems away from equilibrium.

Under this equilibrium assumption it is reasonable to treat interactions between cognitive items as **symmetric**, such that for any pair of cognitive items i, j, the influnece of i on j's behaviour is equal to that of j on i's behaviour. To see why this is the case, suppose that for cognitive items i, j, we have $\omega_{ij} \neq \omega_{ji}$. If the system is at equilibrium, then the probability distribution over states is described by the Boltzmann distribution with Equation 7.54 as its Hamiltonian. Consider the contribution to the Hamiltonian from the interactions between i and j:

$$H_{ij}(\boldsymbol{s}) = -\omega_{ij} s_i s_j - \omega_{ji} s_j s_i \tag{7.59}$$

Since multiplication commutes, $s_i s_j = s_j s_i$, and we may equivalently write this contribution using a single interaction effect:

$$H_{ij}(\boldsymbol{s}) = -(\omega_{ij} + \omega_{ji}) s_i s_j \tag{7.60}$$

Thus when assuming equilibrium there is no representative power to be gained by permitting interaction weights to vary directionally. However, as discussed above, in the present study we are concerned with out-of-equilibrium dynamics, in which the distribution over belief system configurations at time $t + \mathrm{d}t$ is not described by the Boltzmann distribution, but is rather conditional on the configuration observed at time t. In this setting, asymmetric interaction weights do not necessarily have an equivalent symmetric formulation.

For example, consider the simple asymmetric network shown in **REF FIGURE**, comprising two cognitive items, A and B, which are related by a directed edge $A \to B$ with weight 1. Note that this formulation is equivalent to including a directed edge $B \to A$ with weight 0. For simplicity we assume that $\tau_A = \tau_B = 0$.

At time $t + \mathrm{d}t$, for either possible state $s_A \in \{-1, +1\}$, the contribution of A to the system energy is 0, since it has no incoming interaction effects, and has a threshold of zero. On the other hand, B's contribution to the system energy when adopting the state $s \in \{-1, +1\}$ depends on the previous state of A, and is given by $-s^t_A \cdot s$, such that B achieves lower energy when it adopts the previous state of A. Therefore B's behaviour depends on that of A, which in turn is independent of the state of B.

Discuss how this asymmetry should be interpreted.

- In symmetric model: interactions are holding each spin in place.
- In asymmetric model: interactions are reinforcing or damping the future state of other spin.

Introduce formal model

Our model, formally outlined in **REF DEFINITION**, relaxes these assumptions. We redefine interactions as temporal causal effects which positively or negatively reinforce the state of the cognitive item at the head of the relation. We allow for self-interactions, which are interpreted in the same way. Positive self-interactions reflect the *inertia* or *stickiness* of a cognitive item, as seen through a reluctance to change state, with increased interaction weight corresponding to increased inertia. Negative self-interaction weights would reflect unstable cognitive items whose state inherently repels itself in future timesteps. Such negative self-interactions are thus not expected to be observed in reality.

Adjacency matrix makes the asymmetry explicit

7.6 Inverse problem

- Problem with cross-sectional model fitting: show figure of example from above. A's instantaneous behaviour is independent of B's, but B depends on the previous value of A. If we assume asynchronous updating, then perhaps we capture some of this, but not completely accurately. Moreover we cannot infer self-interactions.
- Introduce time-series model fitting.

We first explicate what we mean by the term 'belief system', and then present the formal belief system model which forms the basis for the subsequent chapters.

- Elucidate the core components of a belief system: cognitive items, relations between them
- In a nutshell, what is a belief system?
- Belief systems are individual, but components are likely shared between individuals.
- What is a belief system *not*? i.e., a particular instantiation of beliefs or attitudes.
- What is the nature of the interactions between cognitive items? Why would there be, or not be, a relation between two items?
- What are the dynamics?

Formal model:
- Describe the key components: cognitive items and relations
- Dynamics: maximum entropy model.

A pairwise belief system of size $N \in \mathbb{N}$ is described by a tuple $\mathcal{M} = \langle \boldsymbol{A}, \boldsymbol{J}, \vec{h} \rangle$, where:

- $\boldsymbol{A} \in \{0,1\}^{N \times N}$ is a (possibly directed) adjacency matrix over the N cognitive items, with $A_{ij} = 1 \iff i \to j$,

- $\boldsymbol{J} \in \mathbb{R}^{N \times N}$ describes the weights of the pairwise relations, with the weight of a relation $i \to j$ given by J_{ij}. $J_{ij} = 0 \iff A_{ij} = 0$.

- $\vec{h} \in \mathbb{R}^N$ describes the cognitive item baselines, i.e., the weights of relations with empty tails.

$\mathcal{M}$ confers a conditional probability distribution on the future states of an individual's cognitive items, given their current states. For a cognitive item $i \in [1, N]$, this distribution is:

$$P\left(S_i^{t+1} = s \middle| \boldsymbol{s}^t\right) = \frac{\exp\left[-\frac{1}{T} \cdot -\left(s \cdot h_i + \sum_{j=1}^{N} A_{ji} J_{ji} s_j s\right)\right]}{\sum_{s \in S} \exp\left[-\frac{1}{T} \cdot -\left(s' \cdot h_i + \sum_{j=1}^{N} A_{ji} J_{ji} s_j s'\right)\right]} \tag{7.61}$$

A relation between The relations between cognitive items reflect a tendency for . The cognitive items comprise a collection of N cognitive items which are related in a pairwise faashion (e.g., beliefs or attitudes). A network of pairwise relations

In the forthcoming chapters, we

A belief system of size $N \in \mathbb{N}$ comprises a collection of cognitive items, such as beliefs and attitudes, whose states vary temporally with respect to underlying preferences, external pressures, and other cognitive items. Belief systems are defined at the level of individuals; two individuals within a population may have different belief systems, which exhibit different states.

The notion of a belief system in psychology is open to interpretation. We begin this chapter by explicating our particular interpretation, and clarifying assumptions which underlie our modelling decisions and subsequent analysis. We then present a formal definition for a **computational belief system** which we use in this thesis, and argue that this captures the relevant aspects of the examined phenomena. We additionally discuss and justify our approach to modelling interventions in such computational belief systems.

Finally, we consider practical experimental matters, discussing methods for inferring belief systems from data.

- Introduce chapter: What will we discuss?
- Storytime: What is a belief system? What are the key components, dynamics? What behaviour do they exhibit? What assumptions are we making?
- Modelling belief systems:
  - Common approaches
  - Our assumptions
  - Defining dynamics
  - Interventions
  - Justifying model w.r.t. actual phenomenon.
- Inverse problem: how do we recover belief systems from data?
- Limitations: what is our model incapable of capturing? Where does it differ from phenomenon? What can/can't we infer with our inverse solving approach?

We now present our belief system model. Letting $N \in \mathbb{N}$, a belief system of size N comprises N cognitive items, such as beliefs and attitudes, which vary temporally through interactions with one another and with respect to underlying preferences or external pressures.

Consider a particular cognitive item, σ_i for $i \in [1, N]$, the specific referrent of which is not important. For instance, σ_i might represent an individual's attitude toward travelling or eating meat, emotions such as happiness or anxiety, or a belief, for instance, about tomorrow's weather. Suppose that at time t we measure the state of this cognitive item for a particular individual, and find that $\sigma_i(t) = s$, for $s \in \{-1, +1\}$. We may then reasonably ask what sequence of events has resulted in this state.

- External influences (local fields):
  - Social norms (descriptive, injunctive)
  - Events, experiences (which update knowledge, or affect attitudes)
  - Cost, accessibility, or relevance/context (i.e., states of affairs, can change attitudes)
- Other cognitive items (interactions, adjacency):
  - Beliefs influence beliefs (e.g., implication, overlapping themes)
  - Beliefs influence attitudes (belief that it will be rainy tomorrow $\longrightarrow$ negative attitude).
  - Indirectly: cognitive items affect behaviour (cf. game theory); behaviour reinforces or damps cognitive items, e.g., when enjoyed, or when behaviour causes learning
  - ...?
  - Influences may not be symmetric. For instance, an individual who believes that tomorrow will be rainy may be predisposed to a gloomy mood, but it would be unreasonable to assume that an individual with a gloomy mood is predisposed to particular beliefs about tomorrow's weather. At least not to the same degree of influence. Gloomy moods are multiply-realised, and can fluctuate quickly, often with a time-scale of days or less. Symmetric relations would imply a system in which a bad day could result in a total update to an individual's other beliefs.
- Previous self-state (inertia):
  - Cognitive items may self-reinforce
  - Some are stickier than others, have different natural timescales. e.g., political leanings vs. beliefs about the weather.

7.7 Model description

We now outline the theoretical belief system model adopted in this thesis. In line with existing literature our conceptualisation has its roots in an Ising-style model, operating under the assumption that the temporal dynamics of interest are driven by an individual's efforts — active or otherwise — to reduce cognitive dissonance among their attitudes and beliefs. Our work diverges from existing studies, however, in that we consider the possibility of asymmetric, or directed, interactions between cognitive items.

For $N \in \mathbb{N}$, we conceptualise a belief system of size N as a (possibly directed) network with N nodes. In general we allow for reflexive edges. Each node is considered as referring to a distinct cognitive item. Edges and nodes are weighted with values in $\mathbb{R}$. The sign of an edge's weight describes the tendency (+) or reluctance (−) for the edge's head to align with the tail, with the magnitude specifying the strength of this effect. This captures the notion of cognitive dissonance among items. If two items are connected via the positive edge $A \to B$, and the individual holds different states (for instance $A = -B$), then they are said to experience cognitive dissonance on these items.

Likewise, the sign and magnitude of a node weight describe the general tendency for an individual to exhibit a particular state on a given cognitive item. For instance, if we consider the cognitive item *'attitude toward owning pets'* where a positive state indicates a preference for owning pets, then a negative node weight on this item would indicate that, all else equal, the individual would prefer not to own a pet. We will often refer to node weights as the **baseline** for a cognitive item, or, in the Ising model context, as the **local field effect** on a spin.

With conceptual groundwork now laid, we are in a position to define this model more formally. Letting $N \in \mathbb{N}$, we define a belief system of size N as any tuple $\mathcal{M} = \langle \boldsymbol{A}, \boldsymbol{J}, \vec{h} \rangle$. $\boldsymbol{A} \in \{0,1\}^{N \times N}$ describes the adjacency matrix of a network on N nodes, with $A_{ij} = 1$ if, and only if, there exists a directed edge $i \to j$. $J \in \mathbb{R}^N$ and $\vec{h} \in \mathbb{R}^N$ describe the edge and node weights of the network.

Consider a given spin $i \in [1, N]$. $\mathcal{M}$ imposes a conditional probability distribution on the temporal dynamics of the state of i with respect to the instantaneous states of other spins. Assuming that in general individuals' belief systems tend toward a state of lower cognitive dissonance, i will evolve such that its future state exhibits (at least) no greater dissonance with the individual's baseline tendencies (h_i), nor with the states of their other cognitive items.

$$P\big(S_i^{t+1} = s | \boldsymbol{s}^t\big) = \frac{\exp\Big[-\frac{1}{T} \cdot -\Big(s \cdot h_i + \sum_{j=1}^N A_{ji} J_{ji} s_j s\Big)\Big]}{\sum_{s \in S} \exp\Big[-\frac{1}{T} \cdot -\Big(s' \cdot h_i + \sum_{j=1}^N A_{ji} J_{ji} s_j s'\Big)\Big]} \tag{7.62}$$

Rearrange, pull out the value of s. Remainder is the effective local field. Can show in figures how that varies with different temperatures/effect sizes.

For $S = \{-1, +1\}$, the denominator simplifies to …

For a given spin $i \in [1, N]$, we define the effective local field on i at time t in terms of the instantaneous states of other spins with edges into i.

$$\theta_i(\boldsymbol{s}^t) = h_i + \sum_{j=1}^{N} A_{ji} J_{ji} \cdot s_j \tag{7.63}$$

Through this effective local field, $\mathcal{M}$ imposes a conditional probability distribution on the temporal dynamics of the state of i:

$$P\left(s_i^{t+1} | \boldsymbol{s}^t\right) = \frac{\exp\left[-\frac{1}{T}\right]}{a} \tag{7.64}$$

7.8 Main points

For Kyuri + Vítor:

- Model definition:
  - Generic/Asymmetric: Energy of a configuration
  - Symmetric: Hamiltonian, Boltzmann
  - Glauber dynamics: sequential, parallel

- Model reconstruction:
  - Causal discovery
  - Parameter estimation: cross-sectional likelihood, time-delayed likelihood
  - Add note about observables, why cross-sectional only works for Symmetric without self-loops
  - Assumptions:
    - Same structure for everyone
  - Summarise, with diagram:
    - Causal discovery to restrict structure ($\boldsymbol{A}$)
    - Maximum likelihood to estimate parameters ($\boldsymbol{J}, \vec{h}$)

- Intervention:
  - Method: Change h_i, continue simulation from last measured state
  - Considerations:

 – Model reconstruction assumes homogeneous belief structure among individuals. If some individual has an ‘unstable’ configuration according to this, they will tend toward the population average regardless of intervention. Instead consider behaviour directly after intervention, and look at the difference between intervention behaviour and non-intervention behaviour.

7.8.1 Introducing/motivating our model decisions

Is this the same as the cognitive dissonance framework generally assumed? With asymmetric interactions can we adopt a more general ‘causal’ interpretation?

Still have cognitive dissonance. Consider a belief system which includes the directed relation $a \to b$ with positive weight. If:

- $s_a = 1$, then b updates preferentially toward $s_b = 1$.
- $s_a = -1$, then b updates preferentially toward $s_b = -1$.

(Continue reading from pg 7 in orig. causal discovery book).

7.8.2 Assumptions

Cognitive axes are linearly independent. i.e., the information contained in ‘belief in climate change’ is non-overlapping with ‘belief about the causes of climate change’. The alternative, that they are linearly dependent, could lead to a restriction on possible values. For instance, if two items each depend on a set of lower-level beliefs, a subset of shared low-level beliefs could impose a strict structure on the values that can be obtained. This might not actually be a problem — is this just what we’re modelling?

7.8.3 Belief system dynamics

Let us first explicate what it is that we mean by the term **belief system**. Consider a collection of N cognitive items, comprising beliefs, attitudes (...?).

7.8.4 Configuration energy

Local field energy:

$$H_h(\boldsymbol{s}|\boldsymbol{x}) = -\sum_{i=1}^{N} s_i \cdot h_i(\boldsymbol{x}) \tag{7.65}$$

Where $h_i : \mathbb{R}^N \to \mathbb{R}$ is defined as $\boldsymbol{\alpha}_i^T \boldsymbol{x}$. When no covariates are used this is equivalent to the standard definition of the local field energy.

Interaction energy:

$$H_j(\boldsymbol{s}|\boldsymbol{x}) = -\sum_{\langle ij \rangle} A_{ij} \cdot J_{ij}(\boldsymbol{x}) \cdot s_i s_j \tag{7.66}$$

We take J_{ij} as a constant function of $\boldsymbol{x}$. In other words, the coupling constants do not depend on individual covariates.

Total energy:

$$H(\boldsymbol{s}|\boldsymbol{x}) = H_h(\boldsymbol{s}|\boldsymbol{x}) + H_j(\boldsymbol{s}|\boldsymbol{x}) \tag{7.67}$$

7.8.5 Probability of transition

We first derive the effective local field experienced by s_i at time t. Consider the probability that $s_i^t = 1$, given that $S_{t-1} = \boldsymbol{s}_{t-1}$ Effective local field at spin i:

$$\theta_i(t|\boldsymbol{s}_{t-1}, \boldsymbol{x}_{t-1}) = h_i(\boldsymbol{x}_{t-1}) \tag{7.68}$$

$$P(S_{t+1} = \boldsymbol{s}_{t+1}|S_t = \boldsymbol{s}_t) = \tag{7.69}$$

7.9 Problem statement

- Capture the endogenous dynamics of belief systems, such that we can test the effects of intervention.

7.10 Model definitions

Definition 7.10.1 (Symmetric Ising model)

For $N \in \mathbb{N}$, the Symmetric (network) Ising model of size n comprises N interacting binary variables $s_i \in S$, termed **spins**, where typically S is $\{0, 1\}$ or $\{-1, +1\}$. Formally, the Symmetric Ising model is described by the tuple $M = \langle \boldsymbol{A}, \boldsymbol{J}, \vec{h} \rangle$, where:

- $\boldsymbol{A} \in \{0, 1\}^{N \times N}$ is a symmetric adjacency matrix, describing the connectivity between spins,
- $\boldsymbol{J} \in \mathbb{R}^{N \times N}$ is a symmetric matrix of pairwise spin **interaction constants**, and
- $\vec{h} \in \mathbb{R}^N$ is a vector of **local field constants**.

The **energy** (or **hamiltonian**) of M for a spin configuration $\vec{s} \in S^N$ is given by:

$$H_M(\vec{s}) = -\sum_{i=1}^{N} h_i s_i - \sum_{i=1}^{N} \sum_{j=i}^{n} A_{ij} J_{ij} s_i s_j \tag{7.70}$$

At equilibrium M emits a spin configuration $\vec{s} \in S^N$ with probability described by the Boltzmann distribution:

$$p_M(S^N = \vec{s}) = \frac{\exp\left(-\frac{1}{k_B T} \cdot H_M(\vec{s})\right)}{\sum_{\vec{s}' \in S^N} \exp\left(-\frac{1}{k_B T} \cdot H_M(\vec{s}')\right)} \tag{7.71.1}$$

$$= \frac{1}{Z} \exp\left(-\frac{1}{k_B T} \cdot H_M(\vec{s})\right) \tag{7.71.2}$$

♣

Definition 7.10.2 (Asymmetric Ising model)

For $N \in \mathbb{N}$, the Asymmetric (network) Ising model of size n is relaxation on the Symmetric Ising model definition, which permits $\boldsymbol{A}$ and $\boldsymbol{J}$ to be non-symmetric.

For distinct spins $1 \leq i \neq j \leq N$:

- Interactions may exist in only one direction ($A_{ij} \neq A_{ji}$, or
- The coupling constants may vary directionally ($J_{ij} \neq J_{ji}$).

Self-interactions are identical to those in the Symmetric Ising model, since these are uniquely described by the diagonals of $\boldsymbol{A}$ and $\boldsymbol{J}$.

♣

7.11 Inverse Ising problem

Let $\boldsymbol{Y} \in \{-1, +1\}^{K \times N}$ be a collection of $K \in \mathbb{N}$ observations of $N \in \mathbb{N}$ binary variables, which are assumed to be independent emissions from some unknown Ising model M. The inverse Ising problem consists in identifying a model $\widetilde{M}$ such that $\widetilde{M} \approx M$.

Here we consider the existence and uniqueness of solutions to the inverse Ising problem, and review methods for solving it. We primarily discuss models with configurations in $\{-1, +1\}^N$, though we make clear when the results differ for models over $\{0, 1\}^N$. We first consider Symmetric Ising models with no self-interaction terms (i.e., $A_{ii} = J_{ii} = 0$ for all $i \in [N]$), and then expand more generally to Asymmetric Ising models, and models with self-interaction terms.

7.11.1 Symmetric Ising models without self-interactions

7.11.1.1 Maximum likelihood estimation

The probability that a model M with parameterisation $\boldsymbol{\theta}$ produces the dataset $\boldsymbol{y}$ is more known as the **likelihood**:

$$P(\boldsymbol{Y} = \boldsymbol{y} \mid \boldsymbol{\theta}) = \prod_{\vec{y} \in \boldsymbol{y}} \frac{1}{Z} \exp\left(-\frac{1}{k_B T} H_{M(\vec{y})}\right) \tag{7.72}$$

Maximisation of the likelihood over possible parameterisations is a commonly-used method for solving the inverse Ising problem. In general, even for relatively small N, Equation 7.72 is often a product over many small probabilities, so its calculation is subject to loss in precision due to underflow. As such, we instead typically maximise the **log-likelihood**:

$$\mathcal{L}_{\boldsymbol{y}}(\boldsymbol{\theta}) = \log P(\boldsymbol{Y} = \boldsymbol{y} \mid \boldsymbol{\theta}) \tag{7.73}$$

The optimisation problem is then:

$$\tilde{\boldsymbol{\theta}} = \operatorname{argmax}_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{y}}(\boldsymbol{\theta}) \tag{7.74}$$

For the Symmetric Ising model (with cross-sectional data $\boldsymbol{y}$), the log-likelihood is:

$$\mathcal{L}_{\boldsymbol{y}}(\boldsymbol{\theta}) = \log \prod_{\vec{y} \in \boldsymbol{y}} \frac{1}{Z^{\boldsymbol{\theta}}} \exp\left(-\frac{1}{k_B T} H_M^{\boldsymbol{\theta}}(\vec{y})\right) \tag{7.75.1}$$

$$= -\sum_{\vec{y} \in \boldsymbol{y}} \frac{1}{k_B T} H_M^{\boldsymbol{\theta}}(\vec{y}) + \log\big(Z^{\boldsymbol{\theta}}\big) \tag{7.75.2}$$

- Maybe derive all of this for the more general asymmetric model including self-loops, then come to the conclusion that this is insufficient for models outside the symmetric Ising. Because the log-likelihood derivation should be the same.
- Re-write as sum over possible configurations
- Derive the partial derivatives w.r.t. h_i and J_{ij}

7.11.1.2 Existence

7.11.1.3 Uniqueness

7.12 The Ising model

Definition 7.12.1 (Spin-domain; spin-configuration)

Let S be a set of literals and $n \in \mathbb{N}$. The set S^n comprising all length-n vectors with elements in S is referred to as the as the (S, n)-domain. Where the values of S and n are evident or not relevant, we refer to S^n simply as the *spin-domain*. ♣

Definition 7.12.2 (Spin-configuration)

Let S be a set of literals and $n \in \mathbb{N}$. We refer to an element $\vec{s} \in S^n$ as an (S, n)-configuration. If S and n are evident or not relevant, we instead refer more simply to $\vec{s}$ as a *spin-configuration*. ♣

Example 7.12.3 ((S, n)-domain)

Let $S = \{a, b\}$ and $n = 3$, then the (S, n)-domain comprises the following (S, n)-configurations:

$$\vec{s}_1 = [a, a, a]^T \qquad \vec{s}_2 = [a, a, b]^T \qquad \vec{s}_3 = [a, b, a]^T \qquad \vec{s}_4 = [a, b, b]^T$$
$$\vec{s}_5 = [b, a, a]^T \qquad \vec{s}_6 = [b, a, b]^T \qquad \vec{s}_7 = [b, b, a]^T \qquad \vec{s}_8 = [b, b, b]^T$$

♢

- Introduce the Ising model and its variants in prose, describing the system of spins etc. Refer to other texts for more extensive treatments.
- Then provide formal definitions.

Definition 7.12.4 (Symmetric $(-1, 1)$-Ising model) ♣

Definition 7.12.5 (Kinetic $(-1, +1)$-Ising model) ♣

7.13 Simulation

- Symmetric Ising model at equilibrium: sample using analytic equilibrium probabilities.
  - Show why this works, given detailed balance.
- Non-equilibrium models: Glauber dynamics, parallel glauber dynamics.
  - Give counter-example to detailed balance.

Glauber dynamics new spin state probability:

$$p(s_i(t+1)|\boldsymbol{s}(t)) = \frac{\exp\Big(\frac{1}{T} \cdot s_i(t+1) \cdot \Big[h_i + \sum_{j \neq i} J_{i,j} \cdot s_j(t)\Big]\Big)}{2\cosh\Big(h_i + \sum_{j \neq i} J_{i,j} \cdot s_j(t)\Big)} \tag{7.76}$$

Equivalent form in terms of logistic function:

$$p(s_i(t+1)|\boldsymbol{s}(t)) = \text{logistic}\left(-\frac{2}{T} \cdot s_i(t+1) \cdot \left[h_i + \sum_{j \neq i} J_{i,j} \cdot s_j^t\right]\right) \tag{7.77}$$

7.14 Inverse Ising problem

- What is the inverse Ising problem?
- What is required to solve the inverse Ising problem (existence, uniqueness)?
  - Observables
- Symmetric Ising model: cross-sectional maximum likelihood estimation
- Asymmetric and non-equilibrium models: cross-sectional doesn't work (in general; show conditions for when it does?). Can use time-delayed maximum likelihood estimation. Equivalent to fitting logistic equation for each spin.
- Discuss regularisation?
- Show examples. Including how fit quality varies with number of samples.
- Can only fit self-loops with time-delayed estimation

7.14.1 Constraining allowed interactions

- Causal discovery to: (i) identify likely directional relations between spins, and (ii) prune interactions which are not significant.

Chapter 8

Experiments and results

8.1 Experiments:

8.1.1 Other

Conditional probabilities — how does the specific state of X affect the distribution of Y?

8.1.2 Fitting model

Regularisation:

- Motivate: dealing with small sample size + large parameter count
- Theory: How is it implemented?
- Effects: How does regularisation affect sparsity?
- Parameter choice: Largest λ with EBIC within 2 of minimum EBIC.

The number of estimated parameters is relatively large compared to the number of observations (1700). This is particularly true for the asymmetric model, which features $n(n+1)$ parameters (72 for $n = 8$ beliefs). This can easily result in unreliable parameter estimates with high variance. To reduce the impact of the small sample size on model reliability, we employ a variation of **LASSO** (**L**east **a**bsolute **s**hrinkage and **s**election **p**arameter) regularisation, which penalises non-zero parameter values which do not contribute significantly to the optimal likelihood. We use the following adjusted objective function:

$$f(\boldsymbol{\theta}) = \mathcal{L}_D(\boldsymbol{\theta}) - \lambda \sum_{\theta \in \boldsymbol{\theta}} \sqrt{\theta^2 + \varepsilon} \tag{8.78}$$

where ε is chosen to be small, such that the order of magnitude of $\sqrt{\varepsilon}$ is less than that of the smallest desired value of θ. We take $\varepsilon = 10^{-8}$. The parameter $\lambda \in \mathbb{R}^+$ determines the regularisation strength. Higher values result in sparser models, and $\lambda = 0$ corresponds to no regularisation.

We select the λ which minimises the EBIC (extended bayesian information criterion) for the fit model, while maximising the amount of regularisation. Specifically we select the maximum λ which is close to the minimal EBIC:

$$\lambda = \max\left\{\hat{\lambda} \in \mathbb{R}^+ : \left|\mathrm{EBIC}\left(\hat{\lambda}\right) - \mathrm{EBIC}_{\min}\right| < 2\right\} \tag{8.79}$$

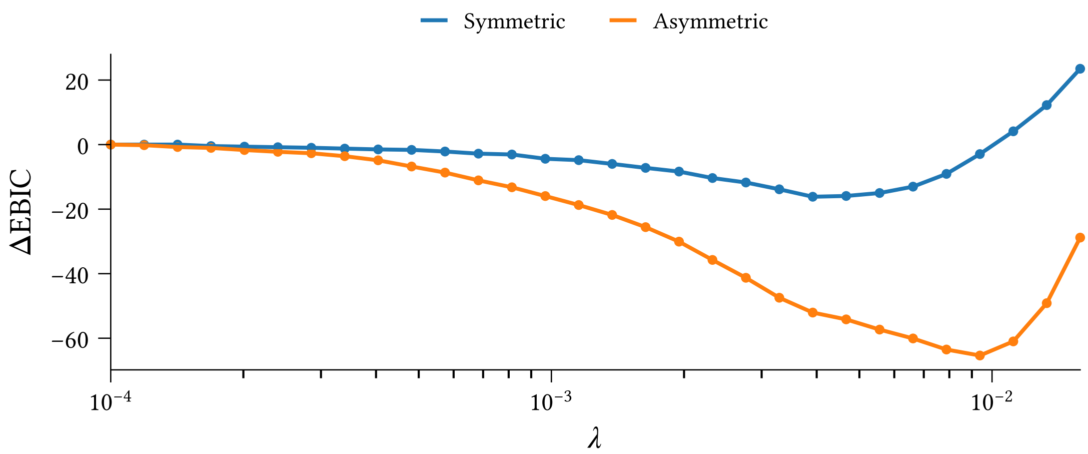

Figure 8.8: TODO

0.009384557359249342

8.1.3 RQ2.1: symmetry or asymmetry of relations in fit models

Fit asymmetric model. Use L1 regularisation to shrink parameters. Use bootstrapping to estimate uncertainty.

Plot directional differences. Characterise relations as symmetric, bidirectional, unidirectional.

8.1.4 RQ2.2: differences between belief systems for different (types of) individuals

Fit separate models based on demographic conditions:
- Male, female
- Urban, suburban, rural

Experiences:
- Extreme weather
- …?

And maybe particular belief states:
- In climate change
- Political stance (dem, rep, indep)

Do for both symmetric and asymmetric. Use L1 regularisation to shrink parameters. Use bootstrapping to estimate uncertainty around inferred parameters. Characterise relations as symmetric, bidirectional, unidirectional.

Consider network features, model features (correlation length, etc.)

8.1.5 RQ3.1: Intervention strategy and effectiveness under sym/asym assumptions

Fit models using bootstrapping. Consider both effects of intervention (how does intervening at S affect other spins) and strategies (how does intervening at different spins affect this *one*). Repeat for varying levels/strengths of intervention.

For strategy, estimate expected ranking of interventions at both collective and individual levels. For the former, measure collective effect, then rank interventions, then take mean across repeats. For the latter, measure expected individual effect, rank interventions, then take mean across individuals.

8.1.6 RQ3.2: Intervention strategy and effectiveness across individuals

Examine distributions of expected outcome effects (per individual) for different interventions. Characterise the individuals found at different parts of the distribution (personas, average magnetisation).

Conditional logic: for a given target belief/attitude, identify the conditions under which each strategy is preferred. Examine this under the distribution implied by the data.

Chapter 9

Discussion

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Chapter 10

Conclusion and future work

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Chapter 11

Ethics and Data Management

A new requirement for the thesis is that there must be a short section in which you reflect on the ethical aspects of your project. This requirement is related to one of the final objectives that a graduated student of the Master of Computational Science must meet: “The graduate of the program has insight into the social significance of Computational Science and the responsibilities of experts in this field within science and in society“. You don’t need to devote an entire chapter to this; a short section or paragraph is sufficient.

I acknowledge that the thesis adheres to the ethical code (https://student.uva.nl/en/topics/ethics-in-research) and research data management policies (https://rdm.uva.nl/en) of UvA and IvI.

The following table lists the data used in this thesis (including source codes). I confirm that the list is complete and the listed data are sufficient to reproduce the results of the thesis. If a prohibitive non-disclosure agreement is in effect at the time of submission “NDA” is written under “Availability” and “License” for the concerned data items.

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